

Xsuite

Lie Groups
Normal Forms
TPSAs

Contents

1	Conventions and Notation	5
1.1	Vectors, matrices, and canonical coordinates	5
1.2	Poisson bracket, Hamiltonian flow, and Lie maps	6
1.3	Point maps and substitution operators	6
1.4	Normal-form phase convention	7
1.5	Series order and external parameters	7
2	Lie Algebra, Lie Groups, and Compositional Maps	9
2.1	Lie Algebra	9
2.1.1	Vector spaces and Lie algebras	9
2.1.2	Lie groups and representations	10
2.1.3	Example: real 2×2 matrices	11
2.1.4	Example: $\mathfrak{su}(N)$	11
2.2	Hamiltonian Maps in Six-Dimensional Phase Space	12
2.2.1	Manifolds	12
2.2.2	Tangent space	12
2.2.3	Symplectic manifolds	14
2.2.4	Symplectic maps	15
2.2.5	Hamiltonian Lie algebra and Poisson algebra	16
2.3	Exponentiation	17
2.3.1	From an infinitesimal generator to a finite transformation . . .	17
2.3.2	Example: $SU(N)$	18
2.3.3	Example: Hamiltonian flow	19
2.3.4	The adjoint representation	20
2.3.5	Lie operators on phase-space functions	21
2.3.6	Why map order can appear to be reversed	22
3	Linear Normal Form	23
3.1	The closed orbit is the origin of the normal form	23
3.2	Symplectic matrices and the eigenvalue problem	24
3.3	Normalisation and rotation do different jobs	26
3.4	Beta functions, phase advances, and tunes	27
3.4.1	One uncoupled degree of freedom	27
3.4.2	Coupled modes and transport around the ring	27
3.5	Phasors and the resonance basis	28
3.6	An implementation with matrices	29

3.7	The same normal form written as maps	30
3.7.1	Conjugating point maps	30
3.7.2	Quadratic generators reproduce the matrices	31
3.8	What the linear normal form contains	32
3.9	Normal forms depending on a parameter	32
3.10	Example: an octupole strength at a fixed location	33
4	Non-Linear Normal Form	35
4.1	The next step after linear normalisation	35
4.1.1	Centre and normalise before expanding	35
4.1.2	Factor the residual map into Lie maps	36
4.2	Solving the Normal-Form Problem Order by Order	37
4.2.1	The resonance basis diagonalises the homological operator . . .	37
4.2.2	Derivation of the homological equation	38
4.2.3	What is removed, and what remains	38
4.3	Geometry of Tori and Dynamics on Tori	39
4.3.1	An action-only normal form	39
4.3.2	Resonant dynamics	40
4.4	Amplitude Detuning	40
4.5	Beta Functions Depending on Amplitude	41
4.5.1	A definition from the fundamental harmonic	41
4.5.2	Mean displacements, higher harmonics, and transport	41
4.6	Momentum Dependence and Nonlinear Chromaticity	42
4.6.1	A family of off-momentum normal forms	42
4.6.2	Chromatic derivatives and mixed terms	42
4.7	A Magnet Strength as an External Parameter	43
4.7.1	Parameter-dependent coordinates and frequencies	43
4.7.2	Example: an octupole at a specified location	44
4.8	Example Implementation in Pseudocode	45
4.9	Validity and Practical Checks	47
5	Generalised TPSAs	49
5.1	TPSAs	49
5.2	Jacobian	50

Chapter 1

Conventions and Notation

The following chapters use several sign conventions and two distinct ways in which a map can act. This short chapter fixes those choices once, so that later formulae can be read without silently changing conventions. The choices are those used throughout this document and are compatible with the accelerator Lie-map framework of Refs. [1, 2, 3, 4].

1.1 Vectors, matrices, and canonical coordinates

Scalars are written in ordinary italic mathematics. Phase-space vectors are written in bold, for example $\mathbf{z}$, and matrices are denoted by capital letters. The superscripts T and $\dagger$ denote transpose and Hermitian conjugation, respectively; i denotes the imaginary unit. Calligraphic symbols such as $\mathcal{M}$ denote maps when this helps distinguish them from their Jacobian matrices. The differential of a map at $\mathbf{z}$ is written $D\mathcal{M}_{\mathbf{z}}$; in a coordinate chart it is represented by the usual Jacobian matrix. Repeated indices are summed only where this is stated or where the Einstein convention is explicitly being used; most sums in the accelerator calculations are written out.

Canonical coordinates are ordered in conjugate pairs,

$$\mathbf{z} = (q_1, p_1, \dots, q_n, p_n)^T. \quad (1.1)$$

With this ordering, define

$$S = \text{diag}(S_2, \dots, S_2), \quad S_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (1.2)$$

In canonical coordinates the symplectic form is therefore

$$\omega = \sum_{j=1}^n dq_j \wedge dp_j, \quad \omega(\mathbf{u}, \mathbf{v}) = \mathbf{u}^T S \mathbf{v}. \quad (1.3)$$

A linear map $\mathbf{z} \mapsto R\mathbf{z}$ is symplectic when $R^T S R = S$; for a nonlinear canonical map the same condition holds for its Jacobian at every phase-space point.

1.2 Poisson bracket, Hamiltonian flow, and Lie maps

The Poisson bracket is defined with the first argument differentiated with respect to q_j in the first term,

$$\llbracket f, g \rrbracket_P = \sum_{j=1}^n \left(\frac{\partial f}{\partial q_j} \frac{\partial g}{\partial p_j} - \frac{\partial f}{\partial p_j} \frac{\partial g}{\partial q_j} \right). \quad (1.4)$$

Thus $\llbracket q_i, p_j \rrbracket = \delta_{ij}$. The Hamiltonian vector field is defined by

$$\iota_{X_f} \omega = df. \quad (1.5)$$

With the symplectic-form convention above, the bracket can equivalently be written

$$\llbracket f, g \rrbracket = \omega(X_f, X_g) = X_g f = -X_f g. \quad (1.6)$$

Thus

$$\dot{q}_j = \frac{\partial H}{\partial p_j}, \quad \dot{p}_j = -\frac{\partial H}{\partial q_j}, \quad \dot{g} = \llbracket g, H \rrbracket. \quad (1.7)$$

The Lie operator places its generator in the **first** Poisson-bracket slot,

$$:f:g \equiv \llbracket f, g \rrbracket. \quad (1.8)$$

Consequently $X_f g = \llbracket g, f \rrbracket = -:f:g$. An expression such as $e^{:f:}$ always means the exponential of the **operator** $:f:$; it is not the scalar function e^f . With these definitions, the forward Hamiltonian point flow is

$$\Phi_H^t(z) = e^{-t:H:} z. \quad (1.9)$$

This minus sign is intentional. It follows from the ordering in Eq. (1.8), rather than from a different sign in Hamilton's equations. In particular, when a Hamiltonian point map is denoted by $\mathcal{E}_H$, this document uses

$$\mathcal{E}_H(z) = e^{-:H:} z. \quad (1.10)$$

Accelerator Lie-map references often write the same finite transformation as $e^{:f:}$ with a **map generator** f . There is no sign conflict: for an autonomous physical Hamiltonian propagated for a parameter interval L , one simply has

$$e^{-L:H:} = e^{:f:}, \quad f = -LH. \quad (1.11)$$

Keeping the physical Hamiltonian H and the Lie-map generator f distinct is useful when comparing conventions across the literature.

1.3 Point maps and substitution operators

A point map $\mathcal{A} : M \rightarrow M$ acts on phase-space points. Its associated substitution operator acts on functions by pullback,

$$\hat{\mathcal{A}}g = g \circ \mathcal{A}. \quad (1.12)$$

If a particle encounters $\mathcal{A}$ first and $\mathcal{B}$ second, then

$$z_{\text{out}} = (\mathcal{B} \circ \mathcal{A})(z_{\text{in}}), \quad \widehat{\mathcal{B} \circ \mathcal{A}} = \widehat{\mathcal{A}} \widehat{\mathcal{B}}. \quad (1.13)$$

Thus point-map composition and substitution-operator multiplication appear in opposite orders. Whenever the distinction matters, this document keeps the composition symbol or the map argument explicit.

1.4 Normal-form phase convention

For each linearly normalised degree of freedom, the action–angle variables are chosen as

$$J_j = \frac{Q_j^2 + P_j^2}{2}, \quad Q_j = \sqrt{2J_j} \cos \phi_j, \quad P_j = -\sqrt{2J_j} \sin \phi_j. \quad (1.14)$$

The corresponding complex phasor is

$$a_j = \frac{Q_j - iP_j}{\sqrt{2}} = \sqrt{J_j} e^{i\phi_j}. \quad (1.15)$$

A positive phase advance μ_j therefore acts as

$$a_j \mapsto e^{i\mu_j} a_j, \quad \phi_j \mapsto \phi_j + \mu_j, \quad \nu_j = \frac{\mu_j}{2\pi}. \quad (1.16)$$

A one-turn eigenvalue determines μ_j modulo 2π and the tune modulo an integer. Continuous phase or tune branches must therefore be chosen when parameters or observation locations are varied.

1.5 Series order and external parameters

A dot denotes differentiation with respect to the continuous flow parameter in context (physical time, longitudinal position, or an auxiliary Lie-flow parameter), whereas an integer subscript n denotes the discrete turn number.

The symbol $\mathcal{O}(\|z\|^r)$ denotes terms of coordinate degree at least r about the stated expansion point. A homogeneous Hamiltonian of degree r generates a first coordinate change of degree $r - 1$, since one Poisson bracket differentiates the generator once. Normal-form calculations are understood locally as formal power-series calculations, or as finite truncations of such series, unless convergence is stated explicitly.

Quantities such as a magnet strength or a prescribed momentum offset may be carried as external Taylor parameters. Unless a parameter is explicitly promoted to a canonical dynamical coordinate with its own conjugate momentum, it is held fixed by the Poisson bracket. Complex phasors are a computational basis; on real physical phase space their conjugates satisfy $a_j^* = \overline{a_j}$.

Chapter 2

Lie Algebra, Lie Groups, and Compositional Maps

Lie methods provide a useful change of viewpoint. Instead of describing a beam-line element only by the map that sends its entrance coordinates to its exit coordinates, one describes an infinitesimal transformation—a **generator**—and obtains the finite map by exponentiation. Linear transfer matrices are the simplest example of this idea. The same language also applies to nonlinear Hamiltonian maps, with the matrix commutator replaced by the Poisson bracket.

2.1 Lie Algebra

2.1.1 Vector spaces and Lie algebras

A vector space V over a field $\mathbb{F}$ (here typically $\mathbb{R}$ or $\mathbb{C}$) is a set whose elements can be added and multiplied by scalars. The result must remain in V , and these operations obey the familiar linear rules: addition is associative and commutative, there is a zero and an additive inverse, and scalar multiplication is associative and distributes over both vector addition and scalar addition. Consequently, if $X, Y \in V$ and $a, b \in \mathbb{F}$, then $aX + bY \in V$. This last closure test is often the most useful quick check in practice.

A **Lie algebra** $\mathfrak{g}$ over $\mathbb{F}$ is a vector space equipped with a binary operation

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \longrightarrow \mathfrak{g}, \quad (X, Y) \longmapsto [X, Y], \quad (2.1)$$

called the **Lie bracket**. For $X, Y, Z \in \mathfrak{g}$ and $a, b \in \mathbb{F}$ it satisfies

$$\begin{aligned} [aX + bY, Z] &= a[X, Z] + b[Y, Z], \\ [Z, aX + bY] &= a[Z, X] + b[Z, Y] \end{aligned} \quad \text{(bilinearity),} \quad (2.2)$$

$$[X, X] = 0 \quad \text{(alternation),} \quad (2.3)$$

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0 \quad \text{(Jacobi identity).} \quad (2.4)$$

Over the fields used here, $\mathbb{R}$ and $\mathbb{C}$, alternation is equivalent to antisymmetry, $[X, Y] = -[Y, X]$. Writing the axiom as alternation is also correct in characteristic 2, where a minus sign does not distinguish antisymmetry from symmetry. The bracket therefore

measures non-commutativity. If every bracket vanishes, the algebra is called abelian. Notice that a Lie algebra has two quite different operations: vector addition and the bracket. The bracket is not in general an associative multiplication.

2.1.2 Lie groups and representations

A **Lie group** G is both a group and a smooth manifold, with smooth group multiplication and inversion. Its Lie algebra is the tangent space at the identity,

$$\mathfrak{g} = T_e G. \quad (2.5)$$

A tangent vector $X \in T_e G$ is the initial velocity of a smooth curve $g(t)$ through the identity: $g(0) = e$ and $\dot{g}(0) = X$. Here t is simply a real curve parameter, defined at least on an open interval around $t = 0$; it need not represent physical time. Different curves with the same initial velocity represent the same tangent vector. Translating this vector from the identity to every other group element produces a left-invariant vector field. The usual commutator of these vector fields, evaluated back at e , defines the Lie bracket. Thus the algebra contains the infinitesimal transformations of the group. It determines the group locally near the identity, but does not by itself contain all global topological information.

The group itself is usually **not** a vector space. Rotation matrices, for example, form a group under matrix multiplication, but the sum of two rotation matrices need not be a rotation matrix, and multiplying a rotation matrix by zero leaves the group. This is precisely why it is useful to move to the tangent space: $T_e G$ is a vector space even when G is curved.

If $\mathfrak{g}$ is finite-dimensional, choose a basis $t_1, \dots, t_n$. These basis elements are commonly called the **generators**. Every infinitesimal transformation is

$$X \in \mathfrak{g} \Leftrightarrow X = x^a t_a, \quad (2.6)$$

where repeated indices are summed, and closure of the bracket gives

$$[t_a, t_b] = f_{ab}^c t_c. \quad (2.7)$$

The numbers f_{ab}^c are the structure constants in this basis. Antisymmetry and the Jacobi identity impose corresponding relations on them. A different basis changes the structure constants but not the Lie algebra. The dimension of $\mathfrak{g}$ equals the dimension of G as a manifold.

A group or algebra can act on many different spaces. A **representation** of a group on a vector space V is a homomorphism¹

$$\rho : G \longrightarrow \mathrm{GL}(V), \quad \rho(g_1 g_2) = \rho(g_1) \rho(g_2), \quad (2.8)$$

and a representation of its algebra is a linear map²

$$\rho_* : \mathfrak{g} \longrightarrow \mathfrak{gl}(V), \quad \rho_*([X, Y]) = \rho_*(X) \rho_*(Y) - \rho_*(Y) \rho_*(X). \quad (2.9)$$

¹The general linear group $\mathrm{GL}_n(\mathbb{F})$ is the set of $n \times n$ invertible matrices over the field $\mathbb{F}$, together with the operation of ordinary matrix multiplication.

²The result on the right is often written as $[\rho_*(X), \rho_*(Y)]$ using the ordinary commutator. One has hence to be careful to distinguish it from the Lie bracket.

A representation is therefore a concrete realisation of abstract generators as matrices or linear operators. It is **faithful** if different algebra elements give different operators. Much of Lie theory in accelerator physics consists of choosing the representation best suited to the calculation: matrices for linear optics, differential operators for nonlinear maps, or finite truncations acting on Taylor coefficients (TPSAs) for numerical implementations of nonlinear maps.

2.1.3 Example: real 2×2 matrices

Consider

$$G = \text{GL}_2(\mathbb{R}) = \{A \in \mathbb{R}^{2 \times 2} : \det A \neq 0\}. \quad (2.10)$$

This is a Lie group under matrix multiplication. It is not a vector space: the zero matrix is excluded, and two invertible matrices can have a singular sum. A curve through the identity can be written $A(t) = I + tX + \mathcal{O}(t^2)$, where X is an arbitrary real 2×2 (potentially singular) matrix. Hence

$$\mathfrak{gl}(2, \mathbb{R}) = \mathbb{R}^{2 \times 2}, \quad [X, Y] = XY - YX. \quad (2.11)$$

The commutator obeys Eqs. (2.2)–(2.4). Using the four matrices E_{ij} , with a one in row i , column j , and zeros elsewhere, as generators gives

$$[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}. \quad (2.12)$$

For example, $[E_{12}, E_{21}] = E_{11} - E_{22}$. This is the elementary model for the same commutator structure encountered with larger transfer matrices.

2.1.4 Example: $\mathfrak{su}(N)$

The special unitary group is

$$\text{SU}(N) = \{U \in \mathbb{C}^{N \times N} : U^\dagger U = I, \det U = 1\}. \quad (2.13)$$

Put $U(t) = I + tX + \mathcal{O}(t^2)$ and differentiate both defining conditions at $t = 0$. One obtains

$$\mathfrak{su}(N) = \{X \in \mathbb{C}^{N \times N} : X^\dagger = -X, \text{tr } X = 0\}, \quad [X, Y] = XY - YX. \quad (2.14)$$

It is a real vector space of dimension $N^2 - 1$. Physicists often prefer Hermitian generators t_a and write $X_a = -it_a$. With the conventional normalisation,

$$[t_a, t_b] = if_{ab}^{c} t_c, \quad [X_a, X_b] = f_{ab}^{c} X_c. \quad (2.15)$$

For $\text{SU}(2)$ one may take $t_a = \sigma_a/2$, where the σ_a are the Pauli matrices, and $f_{ab}^{c} = \epsilon_{abc}$. This example is useful because it separates the abstract real Lie algebra from two common matrix conventions: anti-Hermitian generators are closed under the ordinary commutator, whereas Hermitian generators introduce an explicit factor of i .

2.2 Hamiltonian Maps in Six-Dimensional Phase Space

The application of Lie algebra to nonlinear accelerator physics takes us one step beyond ordinary matrix spaces, and requires a bit of differential geometry to lay the necessary mathematical foundation. One can skip some of these parts and jump directly to the relevant part on the Lie algebra of Hamiltonian maps in Sec. 2.2.5.

2.2.1 Manifolds

A smooth n -dimensional **manifold** M is a space that looks locally like $\mathbb{R}^n$: every point has a neighbourhood that can be described by n coordinates, and changes between overlapping coordinate systems are smooth. Coordinates therefore let us perform calculations locally, but the coordinate tuple assigned to a point should not be confused with the point itself.

For a particle in three degrees of freedom, the familiar canonical phase space is

$$M = \mathbb{R}^6, \quad z = (q_1, p_1, q_2, p_2, q_3, p_3)^T. \quad (2.16)$$

This particular manifold **is** a vector space: componentwise addition and scalar multiplication remain in $\mathbb{R}^6$, and all vector-space axioms follow component by component from those of $\mathbb{R}$. A general manifold, however, need not be a vector space. For instance, the configuration space of a plane pendulum is the circle S^1 , and its phase space is

$$T^*S^1 \simeq S^1 \times \mathbb{R}, \quad (2.17)$$

which is a cylinder: the angle is periodic, whereas its conjugate momentum is real. By itself, the manifold structure supplies neither addition nor scalar multiplication of points. We could define addition as, for instance,

$$(\theta_1, p_1) + (\theta_2, p_2) = (\theta_1 + \theta_2 \pmod{2\pi}, p_1 + p_2), \quad (2.18)$$

which works beautifully. But notice something subtle: here we have implicitly chosen which point of the circle corresponds to $\theta = 0$ to deal with the periodicity of the angle. And this is the key point: the manifold structure by itself supplies no preferred addition law for points. A coordinate-independent addition law can certainly exist, as it does when the manifold is also a Lie group, but it is **additional structure** rather than part of the definition of a manifold. More fundamentally, every finite-dimensional real vector space is topologically equivalent to some $\mathbb{R}^n$ and can be continuously contracted to a point, whereas a loop around the cylinder cannot be contracted. Thus, being locally Euclidean does not make a manifold a vector space.

2.2.2 Tangent space

Although points of a general manifold cannot be added, the infinitesimal directions at any fixed point can. These directions form the **tangent space** $T_p M$ at $p \in M$. One way to define a tangent vector at p is by a smooth curve $\gamma(t)$ through that point, with $\gamma(0) = p$: two curves represent the same tangent vector if their derivatives at

$t = 0$ agree in one (and hence every) local coordinate system. In local coordinates $(x^1, \dots, x^n)$, a tangent vector can be written

$$v = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i} \Big|_p. \quad (2.19)$$

Its components depend on the chosen coordinates, but the vector itself does not. Crucially, $T_p M$ is an n -dimensional vector space, even when M is not: tangent vectors based at the same point may be added and multiplied by scalars. The collection of tangent spaces at all points is the **tangent bundle**

$$TM = \bigsqcup_{p \in M} T_p M, \quad (2.20)$$

$$= \{(p, v) \mid p \in M, v \in T_p M\}. \quad (2.21)$$

The **cotangent space** $T_p^* M$ is the dual vector space of $T_p M$; its elements are linear maps $T_p M \rightarrow \mathbb{R}$. In the coordinates above it has the dual basis $(dx^1, \dots, dx^n)$. Collecting these dual spaces gives the **cotangent bundle**

$$T^* M = \bigsqcup_{p \in M} T_p^* M, \quad (2.22)$$

$$= \{(p, v^*) \mid p \in M, v^* \in T_p^* M\}. \quad (2.23)$$

This explains the notation used for the pendulum: a point of $T^* S^1$ is a pair (θ, p_θ) , where $\theta \in S^1$ specifies the configuration and $p_\theta d\theta \in T_\theta^* S^1$ is its conjugate momentum.

Practical interpretation in phase space. For accelerator calculations, these abstract spaces have a very concrete meaning. If $z = (q_1, p_1, \dots, q_n, p_n) \in M$ is a phase-space point, then a tangent vector $v \in T_z M$ is represented in canonical coordinates by an infinitesimal phase-space variation

$$\delta z = (\delta q_1, \delta p_1, \dots, \delta q_n, \delta p_n)^T. \quad (2.24)$$

Equivalently, it may be the velocity $\dot{z}(0)$ of a curve through z . Thus the tangent vectors of phase space are **not** the canonical momenta: the p_i are already coordinates of the point $z \in M$, whereas a tangent vector describes how the complete phase-space point changes. For example, the Jacobian of a tracking map acts precisely on such variations, $\delta z_{\text{out}} = D\mathcal{M}_z \delta z_{\text{in}}$.

A cotangent vector $\alpha \in T_z^* M$ is instead a linear rule that assigns a number to each infinitesimal variation. In coordinates,

$$\alpha = \alpha_i dz^i, \quad \alpha(v) = \alpha_i v^i. \quad (2.25)$$

The differential of a scalar function is the most common example: $df = (\partial f / \partial z^i) dz^i$ and $df(v) = v^i \partial_i f$, the directional derivative of f along v . Writing a covector as a row vector or “gradient” is therefore a coordinate representation; identifying it with a vector by a dot product additionally uses a metric and should not be confused with the intrinsic vector–covector duality.

There is one related use of the word momentum that is important but lives one level lower. If the phase space itself is a cotangent bundle $M = T^*Q$, then a phase-space point is (q, p) with $q \in Q$ and $p \in T_q^*Q$. In that sense the canonical momenta are the components of a covector on the **configuration space** Q ; they are still not tangent vectors of the phase space M .

2.2.3 Symplectic manifolds

Phase space carries yet another layer of structure. At a point $p \in M$, an alternating covariant 2-tensor is an element

$$\omega_p \in \bigwedge^2 T_p^*M. \quad (2.26)$$

Here $\bigwedge^2 T_p^*M$ is the vector space of antisymmetric bilinear functions

$$\omega_p : T_pM \times T_pM \longrightarrow \mathbb{R}, \quad \omega_p(u, v) = -\omega_p(v, u). \quad (2.27)$$

A **differential 2-form** ω on M is a choice of such a tensor ω_p at every p that varies smoothly with p . The vector space of all differential 2-forms on M is denoted by $\Omega^2(M)$. The pair (M, ω) is a **symplectic manifold** if ω is closed and non-degenerate:

$$d\omega = 0, \quad (2.28)$$

$$\omega_p(v, w) = 0 \quad \forall w \in T_pM \iff v = 0. \quad (2.29)$$

Here $d\omega$ is the exterior derivative of ω , and the second condition is required at every $p \in M$. An antisymmetric bilinear form can be non-degenerate only in even dimensions. Darboux's theorem guarantees that, near every point of a $2n$ -dimensional symplectic manifold, there are canonical coordinates in which

$$\omega = \sum_{j=1}^n dq_j \wedge dp_j. \quad (2.30)$$

For the pendulum this is $\omega = d\theta \wedge dp_\theta$; the same local formula works even though θ is not a global real coordinate. For the special case $M = \mathbb{R}^6$, there is a natural identification

$$T_z\mathbb{R}^6 \simeq \mathbb{R}^6. \quad (2.31)$$

at every z . Therefore, we can write

$$\omega : \mathbb{R}^6 \times \mathbb{R}^6 \rightarrow \mathbb{R}, \quad \omega(u, v) = u^T S v, \quad (2.32)$$

with

$$S = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}. \quad (2.33)$$

We can hence think of S as the coordinate representation of ω .

2.2.4 Symplectic maps

A canonical, or symplectic, map $\mathcal{M} : M \rightarrow M$ is a diffeomorphism that preserves the symplectic form. The easiest way to understand this is to follow two infinitesimal phase-space variations. At $z \in M$, the differential (or **pushforward**)

$$D\mathcal{M}_z : T_z M \longrightarrow T_{\mathcal{M}(z)} M. \quad (2.34)$$

propagates a tangent vector u at the entrance into the tangent vector $D\mathcal{M}_z u$ at the image point. In accelerator language this is the linearised tracking map about the reference point z .

The symplectic form at the image point can therefore be evaluated on the two transported vectors,

$$\omega_{\mathcal{M}(z)}(D\mathcal{M}_z u, D\mathcal{M}_z v). \quad (2.35)$$

This quantity depends on the original point z and on the original vectors $u, v \in T_z M$, so it defines a new 2-form at z . That new form is, by definition, the **pullback** of ω by $\mathcal{M}$:

$$(\mathcal{M}^* \omega)_z(u, v) \equiv \omega_{\mathcal{M}(z)}(D\mathcal{M}_z u, D\mathcal{M}_z v). \quad (2.36)$$

Thus $D\mathcal{M}$ pushes vectors forward, while $\mathcal{M}^*$ pulls forms back so that they can again be compared at the original point. The map is symplectic precisely when this comparison shows no change,

$$\mathcal{M}^* \omega = \omega, \quad (2.37)$$

or, point by point,

$$\omega_{\mathcal{M}(z)}(D\mathcal{M}_z u, D\mathcal{M}_z v) = \omega_z(u, v), \quad (2.38)$$

for every z and every $u, v \in T_z M$. Nothing physically moves “backwards” in a pull-back: it is simply the bookkeeping operation that expresses a form from the output point in terms of input tangent vectors.

In canonical coordinates on the $2n$ -dimensional phase space, let $J(z) : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ be the Jacobian matrix representing $D\mathcal{M}_z$, with entries $J_{ij}(z) = \partial \mathcal{M}_i / \partial z_j$. Thus, if $\mathbf{u}$ is the coordinate vector representing $u \in T_z M$, then $J(z)\mathbf{u}$ represents $D\mathcal{M}_z u \in T_{\mathcal{M}(z)} M$. Since the matrix of ω is the constant matrix S in these coordinates, the preceding identity becomes, for arbitrary coordinate vectors $\mathbf{u}$ and $\mathbf{v}$,

$$(J(z)\mathbf{u})^T S (J(z)\mathbf{v}) = \mathbf{u}^T S \mathbf{v}. \quad (2.39)$$

Equivalently,

$$\mathbf{u}^T (J(z)^T S J(z) - S) \mathbf{v} = 0, \quad (2.40)$$

for all $\mathbf{u}$ and $\mathbf{v}$. The matrix in parentheses must therefore vanish, which gives the familiar Jacobian condition

$$J(z)^T S J(z) = S. \quad (2.41)$$

Consequently, in canonical coordinates, Eq. (2.41) is equivalent to the geometric symplecticity condition (2.37).

For a **linear** map $z \mapsto Rz$, the Jacobian is simply the constant matrix R . The linear symplectic maps therefore form the **symplectic group**

$$\text{Sp}(2n, \mathbb{R}) = \{R \in \text{GL}(2n, \mathbb{R}) \mid R^T S R = S\}. \quad (2.42)$$

This is the group containing the familiar linear transfer matrices of Hamiltonian beam optics. To find its Lie algebra, write a curve through the identity as $R(t) = I + tA + \mathcal{O}(t^2)$ and differentiate $R(t)^T S R(t) = S$ at $t = 0$. One obtains

$$\mathfrak{sp}(2n, \mathbb{R}) = \{A \in \mathbb{R}^{2n \times 2n} \mid A^T S + S A = 0\}. \quad (2.43)$$

Equivalently, SA is symmetric. This provides the direct linear analogue of the nonlinear construction below: a symplectic matrix is a finite canonical transformation, while an element of $\mathfrak{sp}(2n, \mathbb{R})$ is its infinitesimal generator. For quadratic Hamiltonians the Hamiltonian vector field is linear, so exponentiating that vector field reproduces precisely a matrix in $\mathrm{Sp}(2n, \mathbb{R})$. Higher-order Hamiltonians are the nonlinear extension of the same idea.

2.2.5 Hamiltonian Lie algebra and Poisson algebra

The infinitesimal generators of Hamiltonian maps are smooth functions on phase space. Let $C^\infty(M)$ denote all smooth real-valued functions on M . It is a vector space under pointwise operations,

$$(af + bg)(z) = af(z) + bg(z). \quad (2.44)$$

Indeed, a linear combination of smooth functions is smooth, so the set is closed; all the remaining vector-space laws follow from the corresponding laws for real numbers at every point z . This is also an infinite-dimensional vector space. For example, on a canonical coordinate patch the functions $1, q_1, q_1^2, \dots$ (multiplied by a smooth cutoff if the patch is not global) contain arbitrarily large linearly independent subsets.

The relevant functions here are **smooth Hamiltonians**, not necessarily “integrable functions.” An integrable Hamiltonian in the Liouville sense is a Hamiltonian system with sufficiently many independent invariants in involution; such Hamiltonians do not form the vector space used below. Likewise, mere integrability in the measure-theoretic sense would not provide the derivatives needed for a Poisson bracket.

In canonical coordinates the Poisson bracket is

$$\llbracket f, g \rrbracket_P = \sum_{j=1}^n \left(\frac{\partial f}{\partial q_j} \frac{\partial g}{\partial p_j} - \frac{\partial f}{\partial p_j} \frac{\partial g}{\partial q_j} \right). \quad (2.45)$$

This is the convention used by Dragt and Forest: the first argument is differentiated with respect to the coordinate q_j in the first term. In particular,

$$\llbracket q_i, p_j \rrbracket = \delta_{ij}, \quad \llbracket q_i, q_j \rrbracket = \llbracket p_i, p_j \rrbracket = 0. \quad (2.46)$$

It is bilinear and antisymmetric, satisfies the Jacobi identity, and maps two smooth functions to another smooth function. Therefore $(C^\infty(M), \llbracket \cdot, \cdot \rrbracket_P)$ is a Lie algebra. A constant has zero Poisson bracket with every function and generates no motion, so Hamiltonians that differ only by a constant describe the same infinitesimal map. On a connected phase space the non-redundant algebra is consequently

$$\mathfrak{ham}(M) \simeq C^\infty(M)/\mathbb{R}. \quad (2.47)$$

This quotient is again a vector space: define $[f] + [g] = [f + g]$ and $a[f] = [af]$, where $[f]$ denotes all functions that differ from f by a constant. Changing representatives only changes the result by a constant, so these operations are well-defined. The same argument and $\llbracket f + c, g + d \rrbracket = \llbracket f, g \rrbracket$ show that the Poisson bracket is well-defined on the quotient. The algebra remains infinite-dimensional. With the conventions used here, the bracket-preserving identification with the tangent vector fields is

$$[f] \mapsto -X_f, \quad [-X_f, -X_g] = -X_{\llbracket f, g \rrbracket}. \quad (2.48)$$

One may instead identify $[f]$ with X_f . The corresponding function-space bracket then acquires an overall minus sign. This harmless choice is the origin of several sign conventions in the Lie-map literature.

There is a small global qualification behind Eq. (2.47). The tangent algebra to all symplectic maps consists of symplectic vector fields; a symplectic vector field need only be locally Hamiltonian. On the usual $\mathbb{R}^6$ phase space every such field of interest is generated by a global Hamiltonian, while on a manifold with non-trivial topology the Hamiltonian maps can form a proper subgroup of all symplectic maps. For local accelerator maps, $C^\infty(M)/\mathbb{R}$ is the algebra normally used.

The full function space has still more structure. With ordinary pointwise multiplication it is a commutative associative algebra, and the bracket satisfies the Leibniz rule

$$\llbracket f, gh \rrbracket = \llbracket f, g \rrbracket h + g \llbracket f, h \rrbracket. \quad (2.49)$$

A commutative algebra equipped with a Lie bracket satisfying Eq. (2.49) is called a **Poisson algebra**. Thus $C^\infty(M)$ is both a commutative algebra and a Lie algebra, with the two products tied together by the Leibniz rule. Strictly speaking, the quotient $C^\infty(M)/\mathbb{R}$ is a Lie algebra but not a Poisson algebra under pointwise multiplication, because constants are not an ideal for that multiplication. In calculations one therefore keeps representatives in $C^\infty(M)$ and discards irrelevant constant terms when interpreting the generator.

This Poisson-algebra structure is particularly useful for accelerator physics. It permits brackets of polynomial Hamiltonians to be evaluated by ordinary differentiation and multiplication, it extends immediately to Taylor and truncated-power-series representations, and the Leibniz rule controls the action of a generator on products of observables. At the same time, working with a Hamiltonian generator rather than arbitrary Taylor coefficients keeps the underlying map canonical.

2.3 Exponentiation

2.3.1 From an infinitesimal generator to a finite transformation

For a Lie group G with algebra $\mathfrak{g}$, the exponential map

$$\exp : \mathfrak{g} \longrightarrow G. \quad (2.50)$$

is defined through one-parameter subgroups. For every $X \in \mathfrak{g}$ there is a unique one-parameter subgroup $\gamma_X : \mathbb{R} \rightarrow G$ with

$$\gamma_X(0) = e, \quad \dot{\gamma}_X(0) = X, \quad \gamma_X(t+s) = \gamma_X(t)\gamma_X(s). \quad (2.51)$$

The parameter is $t \in \mathbb{R}$, not merely $t \geq 0$: negative values give the inverse transformation, $\gamma_X(-t) = \gamma_X(t)^{-1}$. At this abstract stage t is only a group parameter; depending on the application it may later be physical time, longitudinal position, an angle, a magnet strength multiplied by a length, or simply a dimensionless interpolation parameter. One defines

$$\exp(tX) \equiv \gamma_X(t), \quad \exp(X) = \gamma_X(1). \quad (2.52)$$

Thus exponentiation means following the same infinitesimal generator X for a finite amount of its parameter. This geometric construction is the definition; a power series appears only after choosing a representation in which the generator is a linear operator.

For a matrix Lie group, X is a matrix and the group exponential is exactly the convergent matrix series

$$\exp X = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots. \quad (2.53)$$

More generally, let

$$\rho : G \rightarrow \mathrm{GL}(V). \quad (2.54)$$

be a representation of the group. Its differential at the identity,

$$\rho_* \equiv (d\rho)_e : \mathfrak{g} \rightarrow \mathfrak{gl}(V), \quad (2.55)$$

is the corresponding Lie-algebra representation introduced above. It maps the abstract infinitesimal generator X to the linear operator $\rho_*(X)$ acting on V . Group exponentiation and operator exponentiation then commute with the representation:

$$\rho(\exp tX) = \exp(t\rho_*(X)). \quad (2.56)$$

This equation is not an additional definition: it states that representing a finite group transformation gives the same result as first representing its generator and then exponentiating that operator.

For bounded operators the analogous operator series converges. For unbounded differential operators, or for formal Taylor maps, its domain and convergence require care and the series is often used order by order as a formal series. Moreover, the exponential need not be one-to-one or cover the entire group, although for a finite-dimensional Lie group its image contains a neighbourhood of the identity.

2.3.2 Example: $\mathrm{SU}(N)$

If $X \in \mathfrak{su}(N)$, Eq. (2.53) gives an element of $\mathrm{SU}(N)$. Indeed,

$$(e^X)^\dagger e^X = e^{X^\dagger} e^X = e^{-X} e^X = I, \quad \det(e^X) = e^{\mathrm{tr} X} = 1. \quad (2.57)$$

With Hermitian physics generators this is commonly written

$$U(\boldsymbol{\theta}) = \exp(-i\theta^a T_a). \quad (2.58)$$

The generator is linear in the parameters, while the exponential builds the finite, generally non-commuting transformation.

2.3.3 Example: Hamiltonian flow

The preceding abstract construction becomes especially concrete in phase space. Let $f \in C^\infty(M)$. Its Hamiltonian vector field X_f is the vector field defined, for the symplectic convention used here, by

$$\iota_{X_f}\omega = df, \quad (2.59)$$

where ι_{X_f} denotes insertion of X_f into the first argument of the 2-form. In canonical coordinates this gives

$$X_f = \sum_{j=1}^n \left(\frac{\partial f}{\partial p_j} \frac{\partial}{\partial q_j} - \frac{\partial f}{\partial q_j} \frac{\partial}{\partial p_j} \right), \quad X_f g = \llbracket g, f \rrbracket = -\llbracket f, g \rrbracket. \quad (2.60)$$

Thus $X_f(z) \in T_z M$ gives an infinitesimal direction of motion at every phase-space point.

The **flow** of this vector field is the family of point maps

$$\Phi_f^t : M \rightarrow M. \quad (2.61)$$

defined by the initial-value problem

$$\frac{d}{dt}\Phi_f^t(z) = X_f(\Phi_f^t(z)), \quad \Phi_f^0(z) = z. \quad (2.62)$$

For each initial point z , the curve $t \mapsto \Phi_f^t(z)$ is the integral curve of X_f through z . If $f = H$ is the physical Hamiltonian and t is physical time, Eq. (2.62) is simply Hamilton's equations,

$$\dot{q}_j = \frac{\partial H}{\partial p_j}, \quad \dot{p}_j = -\frac{\partial H}{\partial q_j}, \quad (2.63)$$

or, for any observable g with no explicit t dependence,

$$\frac{dg}{dt} = X_H g = \llbracket g, H \rrbracket. \quad (2.64)$$

In accelerator applications the evolution parameter need not be time; after a suitable Hamiltonian formulation it is often the longitudinal coordinate s . Mathematically nothing changes.

For an autonomous generator, the flow satisfies

$$\Phi_f^{t+s} = \Phi_f^t \circ \Phi_f^s, \quad (\Phi_f^t)^{-1} = \Phi_f^{-t}, \quad (2.65)$$

whenever the relevant solutions exist. It is therefore a one-parameter (local, or global when the vector field is complete) group of diffeomorphisms, and the group exponential is its unit-parameter map,

$$\exp(tX_f) = \Phi_f^t, \quad \exp(X_f) = \Phi_f^1. \quad (2.66)$$

This is the precise meaning of exponentiating a Hamiltonian vector field.

The flow is symplectic. Cartan's formula and Eq. (2.59) give

$$\mathcal{L}_{X_f}\omega = d(\iota_{X_f}\omega) + \iota_{X_f}(d\omega) = d^2f = 0. \quad (2.67)$$

Using the standard identity for differentiation of a pulled-back form along a flow,

$$\frac{d}{dt}(\Phi_f^t)^*\omega = (\Phi_f^t)^*(\mathcal{L}_{X_f}\omega) = 0, \quad (2.68)$$

so $(\Phi_f^t)^*\omega = \omega$ because Φ_f^0 is the identity. In words: infinitesimal Hamiltonian motion preserves the symplectic form, hence so does the finite map obtained by following that motion.

If the generator depends explicitly on the evolution parameter, $f = f_t$, the maps from different parameter values are generated by different vector fields and do not in general form $\exp(tX_f)$ for one fixed X_f . The solution is then a parameter-ordered exponential (or, equivalently, the solution of the non-autonomous flow equation). This is analogous to a sequence of non-commuting beam-line elements, which cannot generally be replaced by the exponential of their simple sum.

2.3.4 The adjoint representation

Every Lie group has a natural representation on its own algebra. Conjugate a group element close to the identity by $g \in G$ and differentiate at the identity:

$$\text{Ad}_g X = \left. \frac{d}{dt} \right|_{t=0} g \exp(tX) g^{-1}. \quad (2.69)$$

This is the **adjoint representation** of the group. For a matrix group,

$$\text{Ad}_g X = gXg^{-1}. \quad (2.70)$$

Differentiating once more gives the adjoint representation of the algebra,

$$\text{ad}_X : \mathfrak{g} \longrightarrow \mathfrak{g}, \quad \text{ad}_X(Y) = [X, Y]. \quad (2.71)$$

The Jacobi identity is exactly the statement

$$[\text{ad}_X, \text{ad}_Y] = \text{ad}_{[X, Y]}. \quad (2.72)$$

Thus the adjoint operators form a Lie algebra under the operator commutator. More precisely, ad_X is an element of $\mathfrak{gl}(\mathfrak{g})$, not in general an element of the original $\mathfrak{g}$; the image $\text{ad}(\mathfrak{g})$ is a Lie subalgebra of operators. Elements in the centre of $\mathfrak{g}$ act as zero, so the adjoint representation is faithful precisely when the centre is trivial.

Exponentiating this representation gives

$$\text{Ad}_{\exp X} = \exp(\text{ad}_X) = I + \text{ad}_X + \frac{\text{ad}_X^2}{2!} + \cdots. \quad (2.73)$$

This is now an ordinary exponential series of linear operators acting on algebra elements. For $\text{SU}(N)$,

$$\text{Ad}_U(Y) = UYU^\dagger, \quad \text{ad}_X(Y) = [X, Y], \quad (2.74)$$

and Eq. (2.73) states $e^X Y e^{-X} = e^{\text{ad}_X} Y$. In the generator basis $X_a = -iT_a$, the matrices of the adjoint representation have components $(\text{ad}_{X_a})_b^c = f_{ab}^c$.

2.3.5 Lie operators on phase-space functions

The adjoint representation of the Hamiltonian algebra is the Lie operator used in non-linear beam dynamics:

$$:f:g \equiv \text{ad}_f(g) = \llbracket f, g \rrbracket. \quad (2.75)$$

It is a linear differential operator on functions, and the Jacobi identity gives

$$[:f:, :g:] = :\llbracket f, g \rrbracket:. \quad (2.76)$$

Constants disappear automatically, so this is a representation of $C^\infty(M)/\mathbb{R}$. The exponentiated Lie operator is

$$e^{:f:}g = g + \llbracket f, g \rrbracket + \frac{1}{2!}\llbracket f, \llbracket f, g \rrbracket \rrbracket + \cdots. \quad (2.77)$$

Unlike the abstract definition of the group exponential, this expression is an exponential power series: it is the operator exponential in the adjoint representation.

The relation to the point flow is now immediate. With the Dragt–Forest convention of Eq. (2.45),

$$:f:g = \llbracket f, g \rrbracket = -X_f g. \quad (2.78)$$

Therefore the operator exponential acts on a function by pullback along the **inverse** Hamiltonian flow,

$$e^{:f:}g = g \circ \Phi_f^{-t} = (\Phi_f^{-t})^* g. \quad (2.79)$$

In particular, applying it componentwise to the coordinate functions gives

$$\Phi_H^t(z) = e^{-t:H:}z. \quad (2.80)$$

This minus sign is not a dynamical convention hidden in Hamilton's equations; it comes solely from defining the Lie operator with the generator in the **first** Poisson-bracket slot, $:f:g = \llbracket f, g \rrbracket$, whereas the Hamiltonian vector field acts as $X_f g = \llbracket g, f \rrbracket$. This is the Dragt–Forest convention used throughout this chapter.

Because of the Leibniz rule, $:f:$ is a derivation. Its exponential is therefore an algebra automorphism:

$$e^{:f:}(gh) = (e^{:f:}g)(e^{:f:}h), \quad \llbracket e^{:f:}g, e^{:f:}h \rrbracket = e^{:f:}\llbracket g, h \rrbracket. \quad (2.81)$$

Acting on the coordinate functions gives a canonical map,

$$\mathcal{M}_f(z) = e^{:f:}z = \Phi_f^{-1}(z), \quad (2.82)$$

where the exponential acts on every component of z and the final equality refers to unit flow parameter. Thus a Lie map with generator f is the inverse unit-parameter flow of the Hamiltonian vector field X_f under the convention adopted here. Products of these exponentials are the **compositional maps** used in Lie factorisations [1, 2]. Non-commutativity is retained through the Baker–Campbell–Hausdorff series; formally,

$$e^{:f:}e^{:g:} = e^{:h:}, \quad h = f + g + \frac{1}{2}\llbracket f, g \rrbracket + \cdots, \quad (2.83)$$

up to an irrelevant additive constant in h .

2.3.6 Why map order can appear to be reversed

Two related objects are frequently denoted by the same symbol in the accelerator literature:

1. a **point map** $\mathcal{A} : M \rightarrow M$, which takes a phase-space point as its argument; and
2. the corresponding **coordinate-substitution operator** $\hat{\mathcal{A}}$, which acts on a function according to

$$\hat{\mathcal{A}}g = g \circ \mathcal{A}. \quad (2.84)$$

Even if $\mathcal{A}$ is nonlinear, $\hat{\mathcal{A}}$ is a linear operator on the vector space of functions. In particular, $\widehat{\mathcal{M}_f} = e^{if}$.

Suppose a particle first passes through $\mathcal{A}$ and then through $\mathcal{B}$. For point maps with explicit arguments,

$$z_1 = \mathcal{A}(z_0), \quad z_2 = \mathcal{B}(z_1) = (\mathcal{B} \circ \mathcal{A})(z_0). \quad (2.85)$$

The rightmost point map acts first. Matrices multiplying a column vector follow the same rule: $z_2 = B A z_0$.

Substitution operators compose in the opposite order:

$$\widehat{\mathcal{B} \circ \mathcal{A}}g = g \circ \mathcal{B} \circ \mathcal{A} = \hat{\mathcal{A}}\hat{\mathcal{B}}g. \quad (2.86)$$

For the coordinate functions themselves this becomes

$$z_2 = (\hat{\mathcal{A}}\hat{\mathcal{B}}z)(z_0). \quad (2.87)$$

The labels in the operator product therefore follow the physical beam-line order from left to right, even though, as operators acting on the object to their right, the rightmost operator is evaluated first. This pullback reversal is the source of the apparently conflicting conventions.

In the notation often associated with Dragt, products of Lie transformations such as

$$e^{if_A} e^{if_B} z. \quad (2.88)$$

are coordinate-substitution operators, so A is encountered before B in the corresponding physical sequence. If instead one writes maps with their phase-space arguments, the same sequence is

$$\mathcal{M}_B(\mathcal{M}_A(z)) = (\mathcal{M}_B \circ \mathcal{M}_A)(z). \quad (2.89)$$

There is no disagreement about composition: one expression acts on functions (including the coordinate functions), while the other acts on points. Whenever an ordering is ambiguous, displaying either the argument or the composition symbol $\circ$ removes it.

Chapter 3

Linear Normal Form

A linear normal form separates two questions about a periodic accelerator: what shape does an oscillation have in physical phase space, and by how much does its phase advance each turn? A canonical change of coordinates answers the first question; a rotation in those coordinates answers the second. The familiar Courant–Snyder construction is the one-degree-of-freedom version of this separation. An eigenvalue problem gives the same construction for coupled motion in several degrees of freedom.

We first derive the normal form with matrices. We then write precisely the same construction with compositional maps and Lie operators, using the conventions of the preceding chapter. This establishes the starting point for the nonlinear normal form: replace the linear change of coordinates by a nonlinear canonical map, and replace constant rotations by more general normal-form dynamics. The map-based approach is developed systematically in Refs. [3, 4].

3.1 The closed orbit is the origin of the normal form

Choose an observation point in the ring and let $\mathcal{M}$ denote its one-turn point map. Use paired canonical coordinates

$$z = (q_1, p_1, \dots, q_d, p_d)^T, \quad z_{n+1} = \mathcal{M}(z_n), \quad (3.1)$$

where n is the turn number. Usually $d = 2$ for transverse motion at fixed momentum, or $d = 3$ when synchrotron motion is included. The closed orbit z_{co} is a fixed point:

$$\mathcal{M}(z_{\text{co}}) = z_{\text{co}}. \quad (3.2)$$

It need not coincide with the zero coordinates of the tracking model. Alignment errors, corrector settings, dispersion, and nonlinear feed-down can all make that distinction essential.

Define the deviation and the one-turn Jacobian by

$$\xi = z - z_{\text{co}}, \quad B = D\mathcal{M}|_{z_{\text{co}}}. \quad (3.3)$$

Taylor expansion about the fixed point gives

$$\mathcal{M}(z_{\text{co}} + \xi) = z_{\text{co}} + B\xi + \mathcal{O}(\|\xi\|^2), \quad \xi_{n+1} = B\xi_n + \mathcal{O}(\|\xi_n\|^2). \quad (3.4)$$

Thus the linear normal form acts on the tangent motion around the closed orbit, rather than on the absolute tracking coordinates. Even a lattice containing nonlinear magnets has such a linearisation; its matrix includes their derivatives evaluated on that orbit.

For an exactly affine map $\mathcal{M}(z) = d + Bz$, the orbit satisfies

$$(I - B)z_{\text{co}} = d. \quad (3.5)$$

For a nonlinear map, the corresponding Newton step is

$$z^{(r+1)} = z^{(r)} + (I - D\mathcal{M}|_{z^{(r)}})^{-1} [\mathcal{M}(z^{(r)}) - z^{(r)}]. \quad (3.6)$$

In code the inverse denotes a linear solve. A unit eigenvalue makes this solve singular and requires a suitable constraint or a reduced problem. For example, without RF there may be no isolated longitudinal fixed point; one normally solves a transverse closed orbit for a prescribed momentum offset instead of applying a three-mode rotation construction blindly.

Subtract the closed orbit before applying the normalisation. If A is the matrix constructed below, the normalised coordinates are $w = A^{-1}(z - z_{\text{co}})$. Using $A^{-1}z$ instead introduces a constant offset, so the apparent actions oscillate even when the linear dynamics is perfectly stable. Computing the Jacobian at the wrong origin can also give incorrect tunes and optics.

3.2 Symplectic matrices and the eigenvalue problem

With the canonical ordering above, introduce

$$S = \text{diag}(S_2, \dots, S_2), \quad S_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad B^T S B = S. \quad (3.7)$$

The last identity follows by differentiating the symplectic one-turn map at its fixed point. It assumes Hamiltonian tracking without damping or other non-symplectic processes. It also assumes genuinely canonical coordinates. A longitudinal tracking variable called δ is not automatically the momentum conjugate to the variable called ζ ; the sign and any required change of variable must be established before using this S in six dimensions.

The normal-mode problem is

$$Bv_j = \lambda_j v_j. \quad (3.8)$$

Because B is real and symplectic, eigenvalues occur with their complex conjugates and reciprocals. For stable elliptic motion they lie on the unit circle and can be written

$$\lambda_j = e^{i\mu_j}, \quad \bar{\lambda}_j = e^{-i\mu_j}. \quad (3.9)$$

The rotation normal form used here requires a complete eigenbasis: the eigenvalues must be semisimple, meaning that their Jordan blocks have size one. Unit-modulus eigenvalues alone are insufficient, since a nontrivial Jordan block produces secular

growth. In the derivation below we take the modes to be distinct and away from $\lambda = \pm 1$. Semisimple degeneracies can also be treated, but require choosing a symplectic basis within the degenerate eigenspace. Hyperbolic modes and non-semisimple cases have other normal forms and do not conserve the positive oscillator actions introduced here.

An ordinary eigensolver leaves the eigenvectors with an arbitrary complex scale. Euclidean normalisation is not the normalisation needed for canonical coordinates. Write

$$\mathbf{v}_j = \mathbf{u}_j + i\mathbf{t}_j, \quad \sigma_j = \frac{\mathbf{v}_j^\dagger S \mathbf{v}_j}{2i} = \mathbf{u}_j^T S \mathbf{t}_j. \quad (3.10)$$

For a simple elliptic eigenvalue away from $\lambda = \pm 1$, σ_j is real and, under the non-degeneracy assumptions used here, nonzero. Its sign is the convention-dependent symplectic (Krein) signature of the chosen member of the conjugate pair. Select the member of each conjugate pair for which $\sigma_j > 0$; if needed replace $(\lambda_j, \mathbf{v}_j)$ by $(\bar{\lambda}_j, \bar{\mathbf{v}}_j)$. Then rescale $\mathbf{v}_j \mapsto \mathbf{v}_j / \sqrt{\sigma_j}$, giving

$$\mathbf{v}_j^\dagger S \mathbf{v}_j = 2i, \quad \mathbf{u}_j^T S \mathbf{t}_j = 1. \quad (3.11)$$

This selection fixes the orientation of each canonical mode, so the phase advance μ_j must be taken from the selected eigenvalue. It need not be the positive principal angle returned by a library.

The symplectic orthogonality of different modes follows directly from the eigenproblem. If $B\mathbf{v} = \lambda\mathbf{v}$ and $B\mathbf{r} = \rho\mathbf{r}$, then

$$\mathbf{v}^T S \mathbf{r} = \mathbf{v}^T B^T S B \mathbf{r} = \lambda \rho \mathbf{v}^T S \mathbf{r}. \quad (3.12)$$

Consequently $\mathbf{v}^T S \mathbf{r} = 0$ when $\lambda \rho \neq 1$. Together with Eq. (3.11), these relations make the real matrix

$$A = (\mathbf{u}_1, \mathbf{t}_1, \dots, \mathbf{u}_d, \mathbf{t}_d) \quad \text{satisfy} \quad A^T S A = S. \quad (3.13)$$

In exact arithmetic no further correction is needed for distinct modes; numerical implementations should check the residual.

Taking real and imaginary parts of the eigenvalue equation gives

$$B\mathbf{u}_j = \cos \mu_j \mathbf{u}_j - \sin \mu_j \mathbf{t}_j, \quad (3.14)$$

$$B\mathbf{t}_j = \sin \mu_j \mathbf{u}_j + \cos \mu_j \mathbf{t}_j. \quad (3.15)$$

Collecting these columns produces the desired decomposition:

$$B = A R A^{-1}, \quad R = \text{diag}(R_2(\mu_1), \dots, R_2(\mu_d)), \quad R_2(\mu) = \begin{pmatrix} \cos \mu & \sin \mu \\ -\sin \mu & \cos \mu \end{pmatrix}. \quad (3.16)$$

The canonical inverse can also be written $A^{-1} = -S A^T S$. This is a matrix derivation of the normal form, and also an explanation of why it is an eigenvalue problem: eigenvalues provide the phase advances, while normalised eigenvectors provide the phase-space basis.

3.3 Normalisation and rotation do different jobs

Define normalised Cartesian coordinates by

$$\mathbf{w} = (Q_1, P_1, \dots, Q_d, P_d)^T = A^{-1}\boldsymbol{\xi}. \quad (3.17)$$

Their linear dynamics is simply $\mathbf{w}_{n+1} = R\mathbf{w}_n$. Each mode therefore has an invariant action

$$J_j = \frac{Q_j^2 + P_j^2}{2}. \quad (3.18)$$

With our orientation of the rotation, write

$$Q_j = \sqrt{2J_j} \cos \phi_j, \quad P_j = -\sqrt{2J_j} \sin \phi_j. \quad (3.19)$$

Then $[\phi_j, J_k] = \delta_{jk}$ and

$$J_{j,n+1} = J_{j,n}, \quad \phi_{j,n+1} = \phi_{j,n} + \mu_j \pmod{2\pi}. \quad (3.20)$$

The phase is undefined at $J_j = 0$, although the Cartesian normal form remains regular there.

The matrix A describes the geometry of the oscillations. It maps normalised circles into physical ellipses and embeds the independent modes in physical coordinates. Its inverse gives the normalised, decoupled coordinates. The matrix R describes the dynamics on those circles. For several nonzero actions, the product of circles is a torus; A embeds that torus in physical phase space and R translates its angles. The complete linear solution is

$$\mathbf{z}_n = \mathbf{z}_{\text{co}} + AR^n A^{-1}(\mathbf{z}_0 - \mathbf{z}_{\text{co}}). \quad (3.21)$$

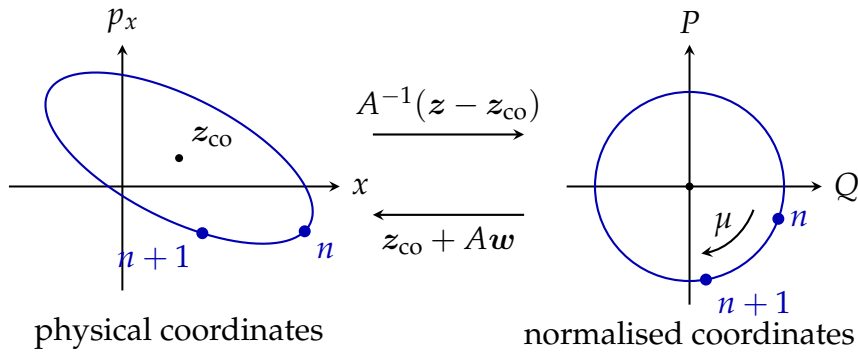


Figure 3.1: The normalisation centres the orbit and changes its ellipse into a circle. The rotation advances the phase on that circle. These are separate parts of the one-turn map.

The normalisation is not unique. Multiplying an eigenvector by $e^{i\theta_j}$ rotates its two real columns, and replacing A by AG with $G = \text{diag}(R_2(\theta_1), \dots, R_2(\theta_d))$ leaves ARA^{-1} unchanged. This is a choice of phase origin, often called a phase gauge. Mode permutations also leave the physical matrix unchanged. A useful convention fixes one

nonzero coordinate component of each eigenvector to be real and positive. In a parameter scan, modes and phases should instead be continued smoothly from the previous point. Such conventions affect individual normal-form coefficients, but not the physical map, actions, or eigenvalues.

3.4 Beta functions, phase advances, and tunes

3.4.1 One uncoupled degree of freedom

For an uncoupled pair (x, p_x) , choose the triangular Courant–Snyder normalisation

$$A_{\text{CS}} = \begin{pmatrix} \sqrt{\beta} & 0 \\ -\alpha/\sqrt{\beta} & 1/\sqrt{\beta} \end{pmatrix}, \quad \begin{pmatrix} Q \\ P \end{pmatrix} = \begin{pmatrix} (x - x_{\text{co}})/\sqrt{\beta} \\ [\alpha(x - x_{\text{co}}) + \beta(p_x - p_{x,\text{co}})]/\sqrt{\beta} \end{pmatrix}. \quad (3.22)$$

Here p_x is the canonical momentum used in the model; identifying it with the geometric slope requires the corresponding paraxial convention. Set $\gamma = (1 + \alpha^2)/\beta$. Direct multiplication of $A_{\text{CS}}R_2(\mu)A_{\text{CS}}^{-1}$ gives

$$B = \begin{pmatrix} \cos \mu + \alpha \sin \mu & \beta \sin \mu \\ -\gamma \sin \mu & \cos \mu - \alpha \sin \mu \end{pmatrix}. \quad (3.23)$$

Thus, away from $\sin \mu = 0$,

$$\cos \mu = \frac{1}{2} \text{tr } B, \quad \beta = \frac{B_{12}}{\sin \mu}, \quad \alpha = \frac{B_{11} - B_{22}}{2 \sin \mu}. \quad (3.24)$$

The sign of $\sin \mu$ is chosen to give $\beta > 0$. The invariant in physical deviations is

$$2J = \gamma(x - x_{\text{co}})^2 + 2\alpha(x - x_{\text{co}})(p_x - p_{x,\text{co}}) + \beta(p_x - p_{x,\text{co}})^2. \quad (3.25)$$

In particular $x - x_{\text{co}} = \sqrt{2\beta J} \cos \phi$. The beta function controls the oscillation amplitude at fixed action; it is a property of the normalisation, whereas the tune is a property of the rotation:

$$\nu_j = \frac{\mu_j}{2\pi}. \quad (3.26)$$

A one-turn eigenvalue fixes ν_j only modulo an integer. Recovering the accumulated tune requires following the phase through the ring or continuing it from a known lattice. Applying $\arccos(\text{tr } B/2)$ alone loses both the orientation and this phase branch.

3.4.2 Coupled modes and transport around the ring

In a coupled lattice, a physical coordinate contains contributions from several normal modes. If row r of A corresponds to a position coordinate x_r , define the projected mode beta functions

$$\beta_{rj} = A_{r,2j-1}^2 + A_{r,2j}^2. \quad (3.27)$$

Indeed,

$$x_r - x_{r,\text{co}} = \sum_{j=1}^d \sqrt{2J_j} [A_{r,2j-1} \cos \phi_j - A_{r,2j} \sin \phi_j], \quad (3.28)$$

$$\left\langle (x_r - x_{r,\text{co}})^2 \right\rangle_\phi = \sum_j \beta_{rj} J_j. \quad (3.29)$$

The average is over independent mode phases. A phase-mixed beam with $\langle J_j \rangle = \epsilon_j$ therefore has $\sigma_{x_r}^2 = \sum_j \beta_{rj} \epsilon_j$ in this linear model. These projected functions are invariant under the phase gauge above. They reduce to the usual beta function for an uncoupled mode, but a coupled lattice generally needs several such functions rather than a single horizontal and a single vertical beta. For a detailed eigenmode treatment see Wolski's formulation of coupled linear optics [5].

Let $L(s_2, s_1)$ be the Jacobian of the point map between two locations, evaluated along their closed orbit. At each location choose a local phase convention for $A(s)$. The transfer matrix has the form

$$L(s_2, s_1) = A(s_2) \text{diag}(R_2(\Delta\psi_1), \dots, R_2(\Delta\psi_d)) A(s_1)^{-1}. \quad (3.30)$$

This expresses the same separation along the whole ring: local optics resides in $A(s)$ and accumulated phase resides in $\Delta\psi_j$. At one full turn, periodic $A(s)$ gives the phases μ_j . Transporting the eigenvectors without removing their accumulating phases is also possible, but then that phase is contained in the columns of A rather than displayed separately in the rotations.

3.5 Phasors and the resonance basis

Complex normalised coordinates diagonalize the rotation itself. Define the phasor and its conjugate by

$$a_j = \frac{Q_j - iP_j}{\sqrt{2}} = \sqrt{J_j} e^{i\phi_j}, \quad \bar{a}_j = \frac{Q_j + iP_j}{\sqrt{2}}, \quad J_j = a_j \bar{a}_j. \quad (3.31)$$

They satisfy

$$[a_j, \bar{a}_k] = i\delta_{jk}, \quad a_{j,n+1} = e^{i\mu_j} a_{j,n}, \quad \bar{a}_{j,n+1} = e^{-i\mu_j} \bar{a}_{j,n}. \quad (3.32)$$

The complex variables are a computational basis; physical phase-space points still obey $\bar{a}_j = \overline{a_j}$.

More generally, for nonnegative multi-indices $\mathbf{k}, \mathbf{l}$, use monomials

$$m_{\mathbf{k}\mathbf{l}} = \prod_j a_j^{k_j} \bar{a}_j^{l_j} = \prod_j J_j^{(k_j+l_j)/2} e^{i(\mathbf{k}-\mathbf{l}) \cdot \boldsymbol{\phi}}. \quad (3.33)$$

Under substitution of the rotation they are eigenfunctions:

$$m_{\mathbf{k}\mathbf{l}} \circ \mathcal{R} = e^{i(\mathbf{k}-\mathbf{l}) \cdot \boldsymbol{\mu}} m_{\mathbf{k}\mathbf{l}}, \quad \mathcal{R}(\mathbf{w}) = R\mathbf{w}. \quad (3.34)$$

This is the **resonance basis**. Unlike real Cartesian monomials, whose coefficients mix under a rotation, each of these monomials only acquires one known phase factor.

Writing $\mathbf{m} = \mathbf{k} - \mathbf{l}$, a discrete-turn resonance occurs when

$$\mathbf{m} \cdot \boldsymbol{\mu} = 2\pi r, \quad \text{or equivalently} \quad \mathbf{m} \cdot \boldsymbol{\nu} = r, \quad r \in \mathbb{Z}. \quad (3.35)$$

Action monomials have $\mathbf{k} = \mathbf{l}$ and are invariant for every tune. Nonzero $\mathbf{m}$ instead describes an angle harmonic. For example a_x^4 has harmonic $4\phi_x$ and becomes invariant at $4\nu_x \in \mathbb{Z}$; $a_x \bar{a}_y$ carries $\phi_x - \phi_y$ and is associated with a difference resonance. A real Hamiltonian contains conjugate pairs of such coefficients.

The resonance condition alone does not make a stable linear oscillator unstable. It identifies harmonics that a perturbation can drive. In the next chapter, removing a nonresonant harmonic requires dividing by a factor proportional to $1 - e^{i\mathbf{m} \cdot \boldsymbol{\mu}}$. This factor vanishes on a resonance and becomes small nearby, explaining both the usefulness of the phasor basis and the need to retain resonant terms in the normal form.

3.6 An implementation with matrices

The following pseudocode assumes distinct elliptic modes and canonical coordinates. The routine `Jacobian` may use derivatives propagated through tracking, automatic differentiation, Taylor maps, or sufficiently accurate finite differences. The map and its Jacobian must use the same observation point, parameters, and coordinate convention.

```
function linear_normal_form(M, initial_orbit, tolerances):
    zco = initial_orbit
    repeat until orbit_residual is small:
        residual = M(zco) - zco
        B = Jacobian(M, zco)
        step = solve(I - B, residual)
        zco = zco + step
    assert norm(M(zco) - zco) < tolerances.orbit

    B = Jacobian(M, zco) # Re-evaluate at converged orbit
    S = block_diagonal([[0, 1], [-1, 0]], number_of_modes)
    assert norm(transpose(B)*S*B - S) < tolerances.symplectic

    eigenvalues, eigenvectors = eig(B)
    pairs = pair_complex_conjugates(eigenvalues, eigenvectors)
    assert stable_semisimple_elliptic_modes(pairs)
    A = empty_real_matrix()
    mu = empty_list()
    for lambda, v in choose_one_member_of_each_pair(pairs):
        sigma = real(conjugate_transpose(v)*S*v / (2*i))
        if sigma < 0:
            lambda, v = conjugate(lambda), conjugate(v)
            sigma = -sigma
        assert sigma > tolerances.mode_norm
```

```

v = v / sqrt(sigma)
v = choose_phase_gauge(v) # Preserve its symplectic norm
append_columns(A, real(v), imag(v))
append(mu, argument(lambda))

A, mu = label_modes_and_continue_phases(A, mu)
R = block_diagonal(rotation(mu[j]) for each mode j)
assert norm(transpose(A)*S*A - S) < tolerances.normalisation
assert norm(B*A - A*R) < tolerances.factorization
return zco, A, mu

function analyze_particle(z, zco, A):
    w = solve(A, z - zco) # Closed-orbit subtraction first
    for each canonical pair Q, P in w:
        J = (Q*Q + P*P)/2
        phi = atan2(-P, Q) # Defined only for nonzero action
        report(J, phi)

function propagate_linear(z, zco, A, mu, turns):
    w = solve(A, z - zco)
    w = block_diagonal(rotation(turns*mu[j]) for each mode j)*w
    return zco + A*w

```

The last two residual checks establish different facts: $A^T S A = S$ checks that the new coordinates are canonical, and $BA = AR$ checks that they reduce the dynamics to rotations. A small eigenvalue residual alone establishes neither the chosen symplectic scale nor the correct mode orientation. For a nonlinear tracking map, the linear propagation agrees locally to first order, and the actions computed with A need not remain exactly constant at finite amplitude.

3.7 The same normal form written as maps

3.7.1 Conjugating point maps

Introduce the translation, linear normalisation, and their composition:

$$\mathcal{T}_{\text{co}}(\xi) = z_{\text{co}} + \xi, \quad \mathcal{L}_A(w) = Aw, \quad \mathcal{C}_0 = \mathcal{T}_{\text{co}} \circ \mathcal{L}_A. \quad (3.36)$$

For each fixed closed orbit the translation is canonical, and $\mathcal{L}_A$ is canonical because $A^T S A = S$. The normalised full point map is

$$\mathcal{F} = \mathcal{L}_A^{-1} \circ \mathcal{T}_{\text{co}}^{-1} \circ \mathcal{M} \circ \mathcal{T}_{\text{co}} \circ \mathcal{L}_A, \quad (3.37)$$

$$\mathcal{F}(w) = A^{-1}[\mathcal{M}(z_{\text{co}} + Aw) - z_{\text{co}}]. \quad (3.38)$$

The rightmost point map acts first. By construction

$$\mathcal{F}(0) = 0, \quad D\mathcal{F}|_0 = R, \quad \mathcal{F}(w) = R w + \mathcal{O}(\|w\|^2). \quad (3.39)$$

For an affine symplectic $\mathcal{M}$, this is exactly $\mathcal{R}$ and

$$\mathcal{M} = \mathcal{C}_0 \circ \mathcal{R} \circ \mathcal{C}_0^{-1}. \quad (3.40)$$

For a nonlinear map the same equation holds for its affine approximation $\mathcal{M}_{\text{lin}}(z) = z_{\text{co}} + B(z - z_{\text{co}})$. Thus matrix similarity has become canonical conjugacy of point maps. The argument $z_{\text{co}} + Aw$ makes explicit both the orbit subtraction and the direction of the normalisation.

For substitution operators $\widehat{\mathcal{U}}g = g \circ \mathcal{U}$, the composition order reverses as in Eq. (2.86). Consequently

$$\widehat{\mathcal{F}} = \widehat{\mathcal{C}}_0 \widehat{\mathcal{M}} \widehat{\mathcal{C}}_0^{-1}, \quad \widehat{\mathcal{M}}_{\text{lin}} = \widehat{\mathcal{C}}_0^{-1} \widehat{\mathcal{R}} \widehat{\mathcal{C}}_0. \quad (3.41)$$

These are identities between operators acting on functions, whereas Eq. (3.40) is an identity between point maps.

3.7.2 Quadratic generators reproduce the matrices

Define the normalised quadratic Hamiltonian

$$H_2(w) = \sum_{j=1}^d \mu_j J_j = \frac{1}{2} w^T D w, \quad D = \text{diag}(\mu_1, \mu_1, \dots, \mu_d, \mu_d). \quad (3.42)$$

Its Hamilton equations are $\dot{Q}_j = \mu_j P_j$, $\dot{P}_j = -\mu_j Q_j$. Its unit-parameter point map is therefore

$$\mathcal{R}(w) = e^{-:H_2:} w, \quad R = \exp(SD), \quad \widehat{\mathcal{R}} = e^{-:H_2:}. \quad (3.43)$$

The minus sign follows from $:H:g = \llbracket H, g \rrbracket$, while Hamiltonian evolution of g is $\llbracket g, H \rrbracket$. For example $:H_2:a_j = -i\mu_j a_j$, so $e^{-:H_2:} a_j = e^{i\mu_j} a_j$, consistent with Eq. (3.31).

In physical deviations the corresponding quadratic Hamiltonian is

$$H_{2,\text{phys}}(\xi) = H_2(A^{-1}\xi) = \frac{1}{2} \xi^T K \xi, \quad K = A^{-T} D A^{-1}, \quad B = \exp(SK). \quad (3.44)$$

Indeed, symplecticity gives $SA^{-T} = AS$, hence $SK = A(SD)A^{-1}$ and the matrix exponentials reproduce Eq. (3.16). This Hamiltonian interpolates the one-turn linear map; it is not generally the local magnet Hamiltonian at the observation point. Its logarithm depends on the phase branches chosen for μ_j .

Translations also have a simple generator. For a constant vector $c = (c_{q_1}, c_{p_1}, \dots)^T$, put

$$H_c(z) = \sum_j (c_{q_j} p_j - c_{p_j} q_j), \quad e^{-:H_c:} z = z + c. \quad (3.45)$$

Taking $c = z_{\text{co}}$ realizes $\mathcal{T}_{\text{co}}$. Likewise, a symplectic matrix sufficiently close to the identity has the form $\exp(SK_A)$ for symmetric K_A , and its point map is generated by $\frac{1}{2} w^T K_A w$. A general real symplectic normalisation can be represented by a product of such quadratic-generated maps; a single real logarithm should not be presumed to exist for every symplectic matrix. The distinction between degree-one translation generators, degree-two linear generators, and higher-degree nonlinear generators is exactly what makes the transition to nonlinear maps systematic.

3.8 What the linear normal form contains

It is useful to separate three layers that are sometimes all called “optics”. The closed orbit z_{co} fixes the origin about which motion is studied. The symplectic matrix A fixes the linear geometry and mode content of infinitesimal oscillations about that orbit. The rotation R fixes their one-turn phase advances. These three objects are already properties of the **full** lattice evaluated at zero oscillation amplitude: nonlinear magnets can affect them through feed-down on an offset closed orbit.

What the linear normal form does not contain is the finite-amplitude effect of higher Taylor coefficients about that orbit. Those coefficients deform the invariant curves or tori, generate amplitude detuning, and drive nonlinear resonances. They are the subject of the next chapter. This separation is particularly important for a pure on-axis octupole: its strength changes the nonlinear normal form while leaving the linearised map unchanged.

3.9 Normal forms depending on a parameter

Let κ be an externally prescribed lattice parameter, such as the integrated strength of one magnet at a fixed position. The problem is now a family $\mathcal{M}_\kappa$, with

$$z_{\text{co}}(\kappa), \quad B(\kappa), \quad A(\kappa), \quad R(\mu(\kappa)). \quad (3.46)$$

Each member must be centred on its own orbit and normalised with its own matrix. Using the reference orbit for a changed lattice leaves a spurious constant term in the normalised map. Here κ is a parameter, rather than an additional canonical coordinate: Poisson brackets and symplecticity refer to the phase-space variables at fixed κ .

Differentiating the fixed-point equation gives

$$(I - B) \frac{dz_{\text{co}}}{d\kappa} = \left. \frac{\partial \mathcal{M}_\kappa}{\partial \kappa} \right|_{z_{\text{co}}}. \quad (3.47)$$

Provided $I - B$ is invertible, this determines the orbit response. The matrix derivative includes the change of expansion point:

$$\frac{dB}{d\kappa} = \partial_\kappa D\mathcal{M}_\kappa|_{z_{\text{co}}} + D^2\mathcal{M}_\kappa|_{z_{\text{co}}} \left[\frac{dz_{\text{co}}}{d\kappa}, \cdot \right]. \quad (3.48)$$

The second term is easily missed: moving the orbit changes the feed-down from nonlinear fields, even at fixed magnet strengths.

For a simple eigenvalue choose a left eigenvector $\ell_j^\dagger$ satisfying $\ell_j^\dagger B = \lambda_j \ell_j^\dagger$ and $\ell_j^\dagger v_j = 1$. Differentiation of the eigenproblem yields

$$\frac{d\lambda_j}{d\kappa} = \ell_j^\dagger \frac{dB}{d\kappa} v_j, \quad \frac{dv_j}{d\kappa} = \frac{1}{2\pi} \text{Im} \left(\lambda_j^{-1} \frac{d\lambda_j}{d\kappa} \right). \quad (3.49)$$

The eigenvector response supplies $dA/d\kappa$ and hence the beta response, for example

$$\frac{d\beta_{rj}}{d\kappa} = 2A_{r,2j-1} \frac{dA_{r,2j-1}}{d\kappa} + 2A_{r,2j} \frac{dA_{r,2j}}{d\kappa}. \quad (3.50)$$

One may obtain these quantities analytically, by Taylor expansion in $\kappa - \kappa_0$, or by a sequence of complete orbit and normal-form calculations. Eigenvectors require a smooth phase gauge and mode labels in all three approaches. At degeneracies the individual mode derivatives require a treatment of the full degenerate subspace.

The same family construction applies to a fixed momentum offset δ_0 when it is conserved by the transverse model. Then $z_{co}(\delta_0)$ includes dispersion and $\nu_j(\delta_0)$ gives off-momentum linear tunes. The derivative $\partial \nu_j / \partial \delta_0$ is linear chromaticity. This is a family of transverse fixed-momentum problems; it is not a replacement for the full longitudinal dynamics when RF makes the momentum oscillate.

3.10 Example: an octupole strength at a fixed location

Place a thin normal octupole at $s = s_0$ and let its integrated strength be κ . Fix the convention with the kick Hamiltonian

$$H_{\text{oct}}(x, y; \kappa) = \frac{\kappa}{24}(x^4 - 6x^2y^2 + y^4), \quad (3.51)$$

where x, y are measured from the magnet's magnetic centre. The associated point map $\mathcal{K}_\kappa = e^{-iH_{\text{oct}}}(x, p_x, y, p_y)^T$ leaves x, y unchanged and gives

$$p_x^+ = p_x^- - \frac{\kappa}{6}(x^3 - 3xy^2), \quad p_y^+ = p_y^- + \frac{\kappa}{6}(3x^2y - y^3). \quad (3.52)$$

If the observation point differs from the octupole, the one-turn map is composed as

$$\mathcal{M}_\kappa = \mathcal{U}_{\text{after}} \circ \mathcal{K}_\kappa \circ \mathcal{U}_{\text{before}}, \quad (3.53)$$

where the two transport maps include the rest of the ring. The location matters because these maps transport the kick through the local optics and phase advance.

Suppose the closed orbit passes through the octupole centre, $x_{co}(s_0) = y_{co}(s_0) = 0$, and the rest of the lattice does not depend on κ . At that orbit the kick is zero and $D\mathcal{K}_\kappa = I$ for every κ . Consequently this ideal thin octupole does not change the closed orbit, the linear one-turn matrix, the zero-amplitude tunes, or the linear beta functions. Its strength is still a useful parameter: it changes the cubic terms of the point map, which generate nonlinear deformation, resonances, and amplitude detuning in the next chapter. Higher-order terms cannot be inferred from B alone.

At an offset orbit x_c, y_c in the magnet, linear feed-down is nonzero. The position Hessian of the Hamiltonian is

$$G_c = \left(\begin{array}{cc} \partial_x^2 H_{\text{oct}} & \partial_x \partial_y H_{\text{oct}} \\ \partial_y \partial_x H_{\text{oct}} & \partial_y^2 H_{\text{oct}} \end{array} \right) \bigg|_{x_c, y_c} = \frac{\kappa}{2} \begin{pmatrix} x_c^2 - y_c^2 & -2x_c y_c \\ -2x_c y_c & y_c^2 - x_c^2 \end{pmatrix}. \quad (3.54)$$

For local deviations $\xi_q = (x - x_c, y - y_c)^T$, $\xi_p = (p_x - p_{x,c}, p_y - p_{y,c})^T$, the linearized kick is

$$\xi_q^+ = \xi_q^-, \quad \xi_p^+ = \xi_p^- - G_c \xi_q^-. \quad (3.55)$$

These grouped vectors only display the kick compactly; the full phase-space ordering remains (x, p_x, y, p_y) . The diagonal entries describe normal quadrupole feed-down

and the off-diagonal entries describe skew quadrupole feed-down. The constant kick also changes the orbit when κ changes, so Eqs. (3.47) and (3.48) must both be used for the full ring response.

For a practical strength scan, run the matrix pseudocode at each κ , propagate each resulting orbit and normalisation to s_o , and continue the mode labels and phases. An on-axis scan provides a useful check: the linear quantities stay constant while the nonlinear Taylor coefficients vary. An offset scan exhibits the feed-down response just derived. Both cases fit the same canonical map framework, whose next step is to normalise the higher-order terms of $\mathcal{F}$ as well as its Jacobian.

Chapter 4

Non-Linear Normal Form

The linear normal form separates the geometry of an oscillation from its phase advance: the matrix A maps circles into phase-space ellipses, while R rotates the normalised coordinates. The nonlinear construction makes the same separation using canonical maps. A nonlinear change of coordinates describes the deformation of the invariant tori, and the transformed one-turn map describes the dynamics on them. Away from resonances, this dynamics is an action-dependent rotation. Near a resonance, some angle-dependent terms must remain and can exchange action between modes.

This chapter develops the construction locally about an elliptic closed orbit. Nonlinear maps and their inverses are understood as formal Taylor series, or as series truncated at a stated order. An action-only normal form at finite order is an approximation to the original map; it does not assert that an arbitrary nonlinear lattice is globally integrable. The map-based viewpoint is also the basis of the generalised Courant–Snyder theory described in Ref. [4].

4.1 The next step after linear normalisation

4.1.1 Centre and normalise before expanding

Let $\mathcal{M}$ be the one-turn point map at a fixed observation location, and let z_{co} be its closed orbit. Use the definitions

$$\mathcal{T}_{\text{co}}(\xi) = z_{\text{co}} + \xi, \quad \mathcal{L}_A(w) = Aw, \quad (4.1)$$

where A is the symplectic linear normalising matrix of the previous chapter. The map in linearly normalised coordinates is

$$\mathcal{F} = \mathcal{L}_A^{-1} \circ \mathcal{T}_{\text{co}}^{-1} \circ \mathcal{M} \circ \mathcal{T}_{\text{co}} \circ \mathcal{L}_A, \quad \mathcal{F}(w) = Rw + \mathcal{O}(\|w\|^2). \quad (4.2)$$

Thus $\mathcal{F}(0) = 0$ and $D\mathcal{F}(0) = R$. In particular, the nonlinear terms describe deviations from the **actual closed orbit**, not deviations from the design trajectory. Expanding about the wrong origin leaves a constant term and gives the wrong linear optics through feed-down. Recentring is equally necessary for every momentum offset or magnet setting considered below.

We seek a canonical near-identity map

$$w = \mathcal{B}(u), \quad \mathcal{B}(0) = 0, \quad D\mathcal{B}(0) = I, \quad (4.3)$$

Here $\mathcal{B}$ is a nonlinear normalising map; it should not be confused with the symbol B used in the preceding chapter for the linear one-turn Jacobian. The matrix A has already removed the linear optical geometry, while $\mathcal{B}$ removes nonresonant nonlinear harmonics order by order.

We require

$$\mathcal{N} = \mathcal{B}^{-1} \circ \mathcal{F} \circ \mathcal{B}. \quad (4.4)$$

The complete transformation from nonlinear normal coordinates to physical coordinates is therefore

$$\mathcal{C} = \mathcal{T}_{\text{co}} \circ \mathcal{L}_A \circ \mathcal{B}, \quad z = \mathcal{C}(u), \quad \mathcal{M} \circ \mathcal{C} = \mathcal{C} \circ \mathcal{N}. \quad (4.5)$$

The last equation is the conjugacy equation. Tracking a normal-coordinate point and then transforming it gives the same result as transforming first and tracking in physical coordinates, to the retained order.

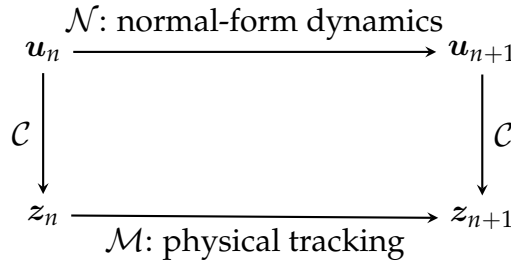


Figure 4.1: The embedding $\mathcal{C}$ describes the physical geometry. The map $\mathcal{N}$ describes the dynamics in normal coordinates. Both paths give the same physical point to the retained order.

4.1.2 Factor the residual map into Lie maps

For clarity, define the Hamiltonian point map

$$\mathcal{E}_H(u) = e^{-:H:} u, \quad \widehat{\mathcal{E}}_H = e^{-:H:}. \quad (4.6)$$

This is the unit-parameter **forward** Hamiltonian flow. The minus sign follows from the convention $:H:g = \llbracket H, g \rrbracket$ in Eq. (2.79); it is essential when extracting tune shifts from a Hamiltonian.

Removing the linear rotation gives $\mathcal{R}^{-1} \circ \mathcal{F} = \text{id} + \mathcal{O}(\|w\|^2)$, where $\mathcal{R}(w) = R w$. In the local canonical neighbourhood used here, the formal logarithm of this near-identity symplectic map is a Hamiltonian vector field (with its Hamiltonian defined up to an irrelevant constant). This is the local setting of the Dragt–Finn Lie factorisation of analytic symplectic maps [1]. We may therefore write, order by order,

$$\mathcal{F} = \mathcal{R} \circ \mathcal{E}_K, \quad K = K_3 + K_4 + K_5 + \dots \quad (4.7)$$

Here K_r is a homogeneous polynomial of degree r in the canonical coordinates. Equivalently, a degree-ordered Lie factorisation is

$$\mathcal{F} = \mathcal{R} \circ \dots \circ \mathcal{E}_{f_5} \circ \mathcal{E}_{f_4} \circ \mathcal{E}_{f_3}, \quad (4.8)$$

$$\widehat{\mathcal{F}} = e^{-:f_3:} e^{-:f_4:} e^{-:f_5:} \dots \widehat{\mathcal{R}}. \quad (4.9)$$

Point-map composition and substitution-operator multiplication have the opposite order, exactly as in Eq. (2.86).

A degree- r Hamiltonian first changes coordinates at degree $r - 1$ because the Poisson bracket differentiates the Hamiltonian once. A Taylor map known through coordinate degree d therefore supplies generators through degree $d + 1$. Cubic generators include sextupole effects, and quartic generators include octupole effects, but these are degrees of the **composed and centred one-turn map**, not labels of individual magnets. For example, brackets of two cubic generators are quartic:

$$\deg \llbracket f_r, g_s \rrbracket = r + s - 2. \quad (4.10)$$

Consequently, sextupoles also contribute to quartic detuning and resonance terms at second order in their strengths.

4.2 Solving the Normal-Form Problem Order by Order

4.2.1 The resonance basis diagonalises the homological operator

Write $\mathbf{u} = (Q_1, P_1, \dots, Q_n, P_n)$ and introduce the phasors

$$a_j = \frac{Q_j - iP_j}{\sqrt{2}} = \sqrt{J_j} e^{i\phi_j}, \quad J_j = a_j a_j^* = \frac{Q_j^2 + P_j^2}{2}. \quad (4.11)$$

With this phase convention, $Q_j = \sqrt{2J_j} \cos \phi_j$, $P_j = -\sqrt{2J_j} \sin \phi_j$ and $\llbracket \phi_j, J_k \rrbracket = \delta_{jk}$. The linear map advances a_j by $e^{i\mu_j}$.

For multi-indices $\mathbf{p}, \mathbf{q}$ of nonnegative integers, use monomials

$$m_{\mathbf{p}\mathbf{q}} = \prod_{j=1}^n a_j^{p_j} (a_j^*)^{q_j} = \prod_{j=1}^n J_j^{(p_j+q_j)/2} e^{i(\mathbf{p}-\mathbf{q}) \cdot \boldsymbol{\phi}}. \quad (4.12)$$

They are eigenfunctions of the substitution operator:

$$\widehat{\mathcal{R}} m_{\mathbf{p}\mathbf{q}} = e^{i\mathbf{k} \cdot \boldsymbol{\mu}} m_{\mathbf{p}\mathbf{q}}, \quad \mathbf{k} = \mathbf{p} - \mathbf{q}. \quad (4.13)$$

The same eigenvalue problem that diagonalised the linear matrix now diagonalises an operator on polynomials. The relevant eigenvalues of $1 - \widehat{\mathcal{R}}$ are $1 - e^{i\mathbf{k} \cdot \boldsymbol{\mu}}$. They vanish at

$$\mathbf{k} \cdot \boldsymbol{\nu} \in \mathbb{Z}, \quad \boldsymbol{\nu} = \frac{\boldsymbol{\mu}}{2\pi}. \quad (4.14)$$

This is a resonance of a **discrete one-turn map**; an integer multiple of 2π is as resonant as zero phase advance.

4.2.2 Derivation of the homological equation

At the first nonlinear order, take $\mathcal{B} = \mathcal{E}_G$ and $\mathcal{F} = \mathcal{R} \circ \mathcal{E}_K$. Pulling back Eq. (4.4) gives

$$\widehat{\mathcal{N}} = e^{-:G:} e^{-:K:} \widehat{\mathcal{R}} e^{:G:}. \quad (4.15)$$

Since $\mathcal{R}$ is canonical,

$$\widehat{\mathcal{R}} : G : \widehat{\mathcal{R}}^{-1} = : G \circ \mathcal{R} :. \quad (4.16)$$

Combining the exponentials to leading order therefore yields

$$\widehat{\mathcal{N}} = e^{-:K+G-G \circ \mathcal{R}+\dots:} \widehat{\mathcal{R}}. \quad (4.17)$$

The omitted Poisson brackets belong to higher polynomial degrees. After lower orders have been normalised, precisely the same calculation applies to the residual degree- r generator K_r .

Let $\mathcal{P}$ project onto the terms that will be retained: always the action terms, and also any selected resonant or near-resonant harmonics. Every exact-resonant term present through the retained order must be included, since its denominator vanishes. Optional near-resonant terms are included together with their conjugate harmonics. Set $H_r = \mathcal{P}K_r$. The degree- r homological equation is

$$(1 - \widehat{\mathcal{R}})G_r = H_r - K_r. \quad (4.18)$$

If $K_r = \sum K_{pq} m_{pq}$, its solution is

$$G_{pq} = \begin{cases} -\frac{K_{pq}}{1 - e^{i(p-q) \cdot \mu}}, & m_{pq} \text{ is removed,} \\ 0, & m_{pq} \text{ is retained.} \end{cases} \quad (4.19)$$

Setting the retained part of G_r to zero fixes a normalisation convention; adding a commuting transformation would otherwise give a different embedding with the same normal-form dynamics. Conjugate coefficients must be paired so that G_r remains real in real coordinates.

Applying $\mathcal{E}_{G_r}^{-1}$ on the left and $\mathcal{E}_{G_r}$ on the right removes the chosen degree- r terms. Recompute the higher-degree residual and repeat. The resulting normalisation is a composition of near-identity canonical maps, rather than a simple sum of the G_r . The Baker–Campbell–Hausdorff brackets generated during these steps are needed for the higher-order coefficients.

4.2.3 What is removed, and what remains

When $p = q$, the monomial is $\prod_j J_j^{p_j}$ and its eigenvalue is one for every tune. These terms cannot be removed by the homological equation. They give the part of the normal-form Hamiltonian that depends only on actions. Away from other resonances, they are the only terms retained.

Terms with $p \neq q$ describe angular harmonics. Their coefficients enter the nonlinear deformation through G_r if they are nonresonant, and remain in the normal-form dynamics if they are resonant. For example, a_x^3 has phase $3\phi_x$ and drives $3\nu_x \in \mathbb{Z}$,

whereas $a_x^2(a_y^*)^2$ has phase $2\phi_x - 2\phi_y$ and drives $2\nu_x - 2\nu_y \in \mathbb{Z}$. Calling a term non-resonant does not mean that its physical effect vanishes: it has been transferred into the coordinate transformation.

The normal form is not unique. The homological equation determines only the part of G_r outside the kernel of $1 - \widehat{\mathcal{R}}$. Adding a commuting generator changes the embedding $\mathcal{B}$ while leaving the normal-form dynamics equivalent. Setting the retained part of G_r to zero, as in Eq. (4.19), is therefore a gauge choice for the normalising transformation, analogous to fixing a phase convention in the linear eigenvectors.

Near a resonance, a small denominator can make the proposed transformation large. Such a term should be retained in a resonant normal form. The decision depends on its strength and the amplitudes of interest, as well as the distance to the resonance; a denominator threshold alone is a numerical policy, not a universal physical criterion.

4.3 Geometry of Tori and Dynamics on Tori

4.3.1 An action-only normal form

If, through the retained polynomial order, the only harmonics in the kernel of $1 - \widehat{\mathcal{R}}$ are the action monomials (and no additional near-resonant harmonics have deliberately been retained), the transformed map is

$$\mathcal{N} = \mathcal{R} \circ \mathcal{E}_{h(\mathbf{J})} = \mathcal{E}_{H_{\text{NF}}}, \quad H_{\text{NF}}(\mathbf{J}) = \sum_j \mu_j J_j + h(\mathbf{J}). \quad (4.20)$$

The two factors commute because both Hamiltonians depend only on actions. Hamilton's equations give

$$J_{j,n+1} = J_{j,n}, \quad (4.21)$$

$$\phi_{j,n+1} = \phi_{j,n} + \Omega_j(\mathbf{J}), \quad \Omega_j = \frac{\partial H_{\text{NF}}}{\partial J_j}. \quad (4.22)$$

These relations hold exactly for the retained action-only normal form and up to the truncation error for the original map. The nonlinear tune is

$$\nu_j(\mathbf{J}) = \frac{\Omega_j(\mathbf{J})}{2\pi}. \quad (4.23)$$

As in the linear case, a one-turn phase fixes the tune modulo an integer; the branch must be followed continuously from the linear optics. The function H_{NF} is an **interpolating Hamiltonian** for the one-turn normal-form map: it generates the same retained one-turn transformation. It is not, in general, the local s -dependent Hamiltonian of the accelerator.

Fixing the positive actions gives a product of circles in normal coordinates, with the angles ϕ locating points on that torus. Its image under $\mathcal{C}$ is a generally deformed torus in physical phase space. If an action vanishes, the corresponding circle collapses and the torus has lower dimension. Approximate invariants in physical coordinates are

$$I_j(z) = J_j(\mathcal{C}^{-1}(z)). \quad (4.24)$$

Computing an action directly from the **linearly** normalised coordinates misses $\mathcal{B}^{-1}$ and generally produces turn-dependent oscillations rather than an invariant.

The map $\mathcal{C}$ specifies the shape, local size, mean displacement, and harmonic content of the oscillation. The Hamiltonian H_{NF} specifies how its phases advance. A term in the embedding can substantially distort an orbit without directly changing its tune at the same perturbative order.

4.3.2 Resonant dynamics

With resonant terms retained, write instead

$$\mathcal{N} = \mathcal{R} \circ \mathcal{E}_{Z(\mathbf{J}, \phi)}, \quad Z = h(\mathbf{J}) + \sum_{\mathbf{k} \neq 0} V_{\mathbf{k}}(\mathbf{J}) e^{i\mathbf{k} \cdot \phi}, \quad (4.25)$$

with conjugate pairs making Z real. During the residual Hamiltonian flow, $\dot{J}_j = -\partial Z / \partial \phi_j$. Individual actions can therefore vary, and the dynamics can develop resonance islands, separatrices, and mode exchange. For one retained resonance vector $\mathbf{k}$, combinations $\mathbf{l} \cdot \mathbf{J}$ with $\mathbf{l} \cdot \mathbf{k} = 0$ are conserved by this ideal normal form, since the angular dependence is only through $\mathbf{k} \cdot \phi$.

At exact resonance the retained harmonics commute with the discrete rotation. Near resonance they generally do not. One may introduce a slow resonant angle and a rotating frame, retaining the tune mismatch alongside the resonant Hamiltonian. Simply replacing the map by $\mathcal{E}_{\sum \mu_j J_j + Z}$ without the required commutation or Baker–Campbell–Hausdorff corrections is not justified.

Thus the interpretation in terms of deformed constant-action tori applies to the non-resonant, action-only approximation. A resonant normal form still separates coordinates from dynamics, but its dynamics is richer than rotation on those tori.

4.4 Amplitude Detuning

Expand the nonresonant Hamiltonian as

$$H_{\text{NF}} = \sum_j \mu_j J_j + \frac{1}{2} \sum_{jk} D_{jk} J_j J_k + \frac{1}{6} \sum_{jkl} E_{jkl} J_j J_k J_l + \cdots, \quad (4.26)$$

where D is symmetric and E is symmetric in its indices. Then

$$\nu_j(\mathbf{J}) = \nu_{j0} + \frac{1}{2\pi} \sum_k D_{jk} J_k + \frac{1}{4\pi} \sum_{kl} E_{jkl} J_k J_l + \cdots. \quad (4.27)$$

The tune footprint is the image of an action domain under this frequency map. Its slope is the amplitude-detuning matrix. Since the frequencies derive from one scalar Hamiltonian,

$$\frac{\partial \nu_j}{\partial J_k} = \frac{\partial \nu_k}{\partial J_j}. \quad (4.28)$$

This relation is a useful consistency check when the actions use the same canonical normalisation.

Quartic action terms produce detuning linear in action, hence quadratic in oscillation amplitude. Sextic terms produce quadratic-in-action detuning. The coefficient is not obtained from the raw quartic kick alone if lower-degree nonlinearities are present: normalising the cubic terms also generates quartic contributions. Similarly, the physical oscillation amplitude is found through $\mathcal{C}$, so it should not be identified with $\sqrt{2J_j}$ in physical units.

4.5 Beta Functions Depending on Amplitude

4.5.1 A definition from the fundamental harmonic

The linear beta function measures how a normalised circle projects into a physical coordinate. At finite amplitude, a deformed torus need not project as an ellipse and contains higher harmonics. There is therefore no single nonlinear beta function that automatically has every property of the linear Courant–Snyder beta. A definition must state which feature of the motion it measures.

For regular nonresonant motion, write the physical embedding at a location s as a Fourier series in the normal-form angles:

$$x(s, \mathbf{J}, \phi) - x_{\text{co}}(s) = \sum_{\mathbf{k} \in \mathbb{Z}^n} X_{\mathbf{k}}(s, \mathbf{J}) e^{i\mathbf{k} \cdot \phi}, \quad X_{-\mathbf{k}} = X_{\mathbf{k}}^*. \quad (4.29)$$

Let e_j be the unit harmonic vector of mode j . A beta function based on its fundamental position harmonic is

$$\beta_{x,j}^{(1)}(s, \mathbf{J}) = \frac{2|X_{e_j}(s, \mathbf{J})|^2}{J_j}. \quad (4.30)$$

The zero-action value is understood as a limit. For uncoupled linear horizontal motion, $X_{e_x} = \sqrt{\beta_x J_x / 2} e^{i\psi_x(s)}$, so Eq. (4.30) reduces to the usual $\beta_x(s)$. For coupled linear motion, it reduces to the projection beta of that normal mode into x .

The nonlinear correction has the form

$$\beta_{x,j}^{(1)}(s, \mathbf{J}) = \beta_{x,j}^{\text{lin}}(s) + \sum_k b_{x,jk}(s) J_k + \dots. \quad (4.31)$$

Its coefficients come from the normalising map $\mathcal{B}$ and the transport of the embedding to s , whereas the detuning coefficients come from H_{NF} . These are different quantities. In particular, the linear relation between beta and local phase advance does not become a general nonlinear identity by substituting an amplitude-dependent beta.

4.5.2 Mean displacements, higher harmonics, and transport

The coefficient X_0 can be nonzero: the mean position of a finite-action torus can differ from the zero-action closed orbit. It is part of the nonlinear embedding and should not be mistaken for an error in the closed-orbit subtraction. Higher X_k describe additional distortion. In an uncoupled one-mode example, an alternative measure is

$$\beta_x^{\text{rms}}(J) = \frac{\langle (x - \langle x \rangle)^2 \rangle_\phi}{J} = \frac{1}{J} \sum_{k \neq 0} |X_k(J)|^2. \quad (4.32)$$

It agrees with the linear beta, but includes all harmonics and generally differs from $\beta_x^{(1)}$ at finite amplitude.

If $\mathcal{P}_{s \leftarrow s_0}$ transports physical points from the normal-form observation point s_0 to s , the torus embedding there is

$$\mathcal{C}_s = \mathcal{P}_{s \leftarrow s_0} \circ \mathcal{C}_{s_0}. \quad (4.33)$$

A local phase convention can subsequently be chosen for the normal coordinates. Extracting Fourier coefficients from this transported map gives amplitude-dependent optics along the lattice. Fitting a single ellipse to a strongly distorted projection does not in general reproduce these harmonic beta functions.

4.6 Momentum Dependence and Nonlinear Chromaticity

4.6.1 A family of off-momentum normal forms

Suppose the relative momentum offset δ_0 is constant during the motion, as in a transverse coasting-beam model. It then labels a family of four-dimensional symplectic maps,

$$\mathcal{M}_{\delta_0}, \quad z_{\text{co}}(\delta_0), \quad A(\delta_0), \quad \mathcal{B}_{\delta_0}, \quad H_{\text{NF}}(\mathbf{J}; \delta_0). \quad (4.34)$$

Both the coordinates and the dynamics depend on momentum. One first solves for the off-momentum closed orbit, including its nonlinear dispersion, and then normalises about that orbit. Subtracting only the on-momentum orbit, or only its linear dispersion correction, is insufficient for a higher-order chromatic calculation.

Use the same canonical coordinate and momentum convention throughout the family. In particular, comparisons at fixed action require a consistent normalisation; holding a physical launch displacement fixed generally changes the action when A or $\mathcal{B}$ changes.

The nonresonant Hamiltonian becomes

$$H_{\text{NF}}(\mathbf{J}; \delta_0) = \sum_j 2\pi v_{j0}(\delta_0) J_j + \frac{1}{2} \sum_{jk} D_{jk}(\delta_0) J_j J_k + \dots \quad (4.35)$$

Consequently,

$$v_j(\mathbf{J}; \delta_0) = v_{j0}(\delta_0) + \frac{1}{2\pi} \sum_k D_{jk}(\delta_0) J_k + \dots \quad (4.36)$$

The Fourier beta functions also depend on δ_0 , through $\mathcal{C}_s(\mathbf{J}, \phi; \delta_0)$.

4.6.2 Chromatic derivatives and mixed terms

Define the chromaticity of order r by derivatives, so the factorial convention is explicit:

$$\tilde{\zeta}_j^{(r)}(\mathbf{J}) = \left. \frac{\partial^r v_j(\mathbf{J}; \delta_0)}{\partial \delta_0^r} \right|_{\delta_0=0}, \quad (4.37)$$

$$v_j(\mathbf{J}; \delta_0) = v_j(\mathbf{J}; 0) + \sum_{r \geq 1} \frac{\tilde{\zeta}_j^{(r)}(\mathbf{J})}{r!} \delta_0^r. \quad (4.38)$$

At zero action, $\tilde{\zeta}_j^{(1)}(0)$ is the usual linear chromaticity. The derivatives with $r \geq 2$ describe nonlinear chromaticity of the small-amplitude tune. At finite action, even the first chromatic derivative can depend on amplitude:

$$\tilde{\zeta}_j^{(1)}(\mathbf{J}) = \tilde{\zeta}_j^{(1)}(0) + \frac{1}{2\pi} \sum_k D'_{jk}(0) J_k + \cdots. \quad (4.39)$$

Thus nonlinear chromaticity and amplitude-dependent chromaticity are related but distinct effects. Mixed terms such as $J_j J_k \delta_0$ and $J_j \delta_0^2$ in the Hamiltonian retain both kinds of information.

If RF motion makes momentum dynamical, δ cannot be treated as a constant external label. The appropriate construction uses all three canonical pairs about a six-dimensional periodic orbit. Its nonresonant frequencies depend on the longitudinal action J_s , and its resonance basis includes $k_x \nu_x + k_y \nu_y + k_s \nu_s \in \mathbb{Z}$. A scan in constant δ_0 is useful transverse optics information, but it does not itself describe synchrotron modulation or synchrotron resonances. The longitudinal coordinates must also be a canonical pair in the adopted tracking convention.

4.7 A Magnet Strength as an External Parameter

4.7.1 Parameter-dependent coordinates and frequencies

Let λ be a magnet strength or another lattice knob. The entire construction becomes a family:

$$\mathcal{M}_{\delta_0, \lambda} = \mathcal{C}_{\delta_0, \lambda} \circ \mathcal{N}_{\delta_0, \lambda} \circ \mathcal{C}_{\delta_0, \lambda}^{-1}. \quad (4.40)$$

For example, with $\eta = \lambda - \lambda_*$ one may expand

$$D_{jk}(\delta_0, \lambda) = D_{jk,*} + D_{jk,\delta} \delta_0 + D_{jk,\lambda} \eta + D_{jk,\delta\lambda} \delta_0 \eta + \cdots. \quad (4.41)$$

The embedding has its own expansion. A knob can therefore change both the tune footprint and the amplitude-dependent optics.

External parameters are not canonical dynamical variables: they have no conjugate momentum in this calculation and do not enter the transverse Poisson bracket. Taylor coefficients in these parameters can be carried through the map operations as well. The normal-form sequence follows polynomial degree in the dynamical coordinates; parameter degree can be truncated separately. The resonance denominators, eigenvectors, and closed orbit must also be expanded or differentiated, rather than held fixed without justification.

Choose mode labels, eigenvector phases, and tune branches continuously along the parameter family. If a selected denominator crosses a resonance, a nonresonant normalisation cannot be differentiated smoothly through that crossing; retain the relevant harmonic or use a different local expansion.

4.7.2 Example: an octupole at a specified location

Consider a thin normal octupole at s_o , with integrated strength $\kappa = \kappa_* + \lambda$ defined by the Hamiltonian

$$H_o(x, y) = \frac{\kappa}{24}(x^4 - 6x^2y^2 + y^4). \quad (4.42)$$

Its point kick leaves x, y fixed and gives

$$\Delta p_x = -\frac{\kappa}{6}(x^3 - 3xy^2), \quad \Delta p_y = \frac{\kappa}{6}(3x^2y - y^3). \quad (4.43)$$

The Hamiltonian and the strength use the same canonical momentum convention as the lattice map. This explicit definition avoids factorial and sign ambiguities in multipole conventions.

First assume an uncoupled linear lattice, a closed orbit through the octupole centre, no other nonlinearities, and weak κ , away from the relevant resonances. At the octupole,

$$x = \sqrt{2\beta_x(s_o)J_x} \cos \theta_x, \quad y = \sqrt{2\beta_y(s_o)J_y} \cos \theta_y. \quad (4.44)$$

The local angles include the phase advance from the observation point. The quartic action part is the angle average of H_o :

$$\langle x^4 \rangle = \frac{3}{2}\beta_x^2 J_x^2, \quad \langle x^2 y^2 \rangle = \beta_x \beta_y J_x J_y, \quad (4.45)$$

$$h_o = \kappa \left(\frac{\beta_x^2 J_x^2}{16} - \frac{\beta_x \beta_y J_x J_y}{4} + \frac{\beta_y^2 J_y^2}{16} \right). \quad (4.46)$$

Here the beta functions are evaluated at s_o . Differentiating gives the first-order tune shifts

$$\Delta \nu_x = \frac{\kappa}{16\pi} \beta_x^2 J_x - \frac{\kappa}{8\pi} \beta_x \beta_y J_y, \quad (4.47)$$

$$\Delta \nu_y = \frac{\kappa}{16\pi} \beta_y^2 J_y - \frac{\kappa}{8\pi} \beta_x \beta_y J_x. \quad (4.48)$$

For this example, $\partial \Delta \nu_j / \partial \lambda$ is obtained by replacing κ with one. An on-axis octupole changes the finite-action tunes at this order while leaving the zero-action linear tunes unchanged.

The same octupole also deforms the tori. To see both outputs explicitly, set $J_y = 0$, take the observation point immediately before the kick, and write the remainder of the linear turn as $\mathcal{R}$. Then $\mathcal{F} = \mathcal{R} \circ \mathcal{E}_{K_4}$ with

$$K_4 = \kappa \beta_x^2 J_x^2 \left(\frac{1}{16} + \frac{\cos 2\phi_x}{12} + \frac{\cos 4\phi_x}{48} \right). \quad (4.49)$$

The constant harmonic is h_o . Away from the second- and fourth-order resonances, the other two harmonics give

$$G_4 = \frac{\kappa \beta_x^2 J_x^2}{24 \sin \mu_x} \sin(2\phi_x - \mu_x) + \frac{\kappa \beta_x^2 J_x^2}{96 \sin 2\mu_x} \sin(4\phi_x - 2\mu_x). \quad (4.50)$$

Each harmonic solves $G_4(\phi_x) - G_4(\phi_x + \mu_x) = h_o - K_4$. The map $\mathcal{B} = \mathcal{E}_{G_4}$ gives

$$\mathbf{w} = \mathbf{u} - \llbracket G_4, \mathbf{u} \rrbracket + \mathcal{O}(\kappa^2), \quad (4.51)$$

at this perturbative order. The fundamental position harmonic yields, at this particular observation point and with this phase convention,

$$\beta_x^{(1)}(J_x) = \beta_x - \frac{\kappa \beta_x^3}{8} \cot \mu_x J_x + \mathcal{O}(\kappa^2), \quad (4.52)$$

at fixed sufficiently small J_x . This comes from the $2\phi_x$ term in G_4 ; other harmonics affect the orbit shape as well. Transporting the embedding gives the beta correction at other locations. The detuning and this beta correction therefore arise from different parts of the **same** normal-form calculation.

In two planes the nonconstant harmonics include $4\phi_x$, $4\phi_y$, $2\phi_x$, $2\phi_y$, and $2\phi_x \pm 2\phi_y$. Their coefficients acquire phase factors when transported to a different observation point. Contributions from several octupoles add as phasors for a specified resonance, so their relative locations matter. Their first-order action averages instead add without these phase factors, weighted by the local beta functions.

An off-axis orbit produces feed-down. For example, take $x_{\text{co}}(s_o, \delta_0) = D_x(s_o)\delta_0 + \mathcal{O}(\delta_0^2)$ and $y_{\text{co}} = 0$. The quadratic Hamiltonian about that orbit contains the effective integrated quadrupole strength

$$\Delta k_1 = \frac{\kappa}{2} D_x^2 \delta_0^2 + \mathcal{O}(\delta_0^3). \quad (4.53)$$

To first order in κ , its small-amplitude tune contribution is

$$\Delta \nu_{x0} = \frac{\kappa \beta_x D_x^2}{8\pi} \delta_0^2 + \mathcal{O}(\delta_0^3), \quad \Delta \nu_{y0} = -\frac{\kappa \beta_y D_x^2}{8\pi} \delta_0^2 + \mathcal{O}(\delta_0^3). \quad (4.54)$$

Here the beta and dispersion are their on-momentum values at s_o . Thus this feed-down gives a second chromatic derivative even though the on-axis, on-momentum linear tune response vanishes. Momentum dependence of the beta functions and of the strength convention also changes the detuning coefficients in Eq. (4.47), producing mixed amplitude–chromatic effects. In a lattice with existing sextupoles, offsets, or coupling, the full centred map includes additional terms and must be normalised rather than using these isolated-octupole formulae unchanged.

4.8 Example Implementation in Pseudocode

The algorithm below assumes access to Taylor maps, composition, inversion, and Poisson brackets. A TPSA implementation is one way to supply these operations; the following chapter describes that representation. Every map operation is truncated to coordinate degree d , and Hamiltonians are kept through degree $d + 1$. The resonance selection is fixed for the local parameter domain. All compositions below are compositions of **point maps**, with the rightmost map acting first. The Taylor representation of $\mathcal{M}$ takes absolute physical coordinates as arguments, even though it is expanded about the closed orbit. The routine `symplectic_eigenbasis` performs the eigenvector part of the preceding chapter’s matrix algorithm.

```

input: one_turn_map, coordinate_degree d, parameter_orders
      fixed local set of retained resonance harmonics

co = solve_closed_orbit(one_turn_map, delta0, lambda)
M = taylor_map(one_turn_map, around=co,
               coordinate_degree=d, parameter_orders=parameter_orders)
A, mu = symplectic_eigenbasis(jacobian(M, co))
T = translation(co)
L = linear_map(A)
F = inverse(L) o inverse(T) o M o T o L
R = rotation_map(mu)

current = F
B = identity_map()
Z = 0                                     # retained nonlinear Hamiltonian

for r = 3, ..., d + 1:
    # All lower degrees have already been normalised.
    residual = inverse(E(Z)) o inverse(R) o current
    V = homogeneous_part(residual - identity_map(), r - 1)
    K_r = -dot(u, S * V) / r # V = S * gradient(K_r)
    check_gradient(K_r, -S * V)

    Kc = to_phasor_basis(K_r)
    Hc = 0
    Gc = 0
    for coefficient c, monomial a**p * conjugate(a)**q in Kc:
        k = p - q
        if k == 0 or harmonic_is_retained(k):
            Hc += c * monomial
        else:
            denom = 1 - exp(i * dot(k, mu))
            check_small_denominator(c, denom, action_domain)
            Gc += (-c / denom) * monomial

    H_r = to_real_canonical_coordinates(Hc)
    G_r = to_real_canonical_coordinates(Gc)
    check_reality(H_r, G_r)
    step = E(G_r) # E(H)(u) = exp(-:H:) u
    current = inverse(step) o current o step
    B = B o step
    Z = Z + H_r
    # Recomputed residuals include all higher-order brackets.

```

The generator extraction uses Euler's identity for a homogeneous polynomial: if $V_{r-1} = S \nabla K_r$, then $K_r = -\mathbf{u}^T S V_{r-1} / r$. The gradient check detects a residual that is not Hamiltonian to the required order. In the residual above, $\mathcal{E}(Z)^{-1}$ removes the already retained lower-order flow; brackets with a new degree- r term affect higher degrees and are picked up on subsequent iterations. All exponentials, including that

of Z , must be evaluated with the same truncation rules.

If parameters are represented by Taylor series, the closed-orbit solve, linear normalisation, and resonance denominators are performed with those series as coefficients. Polynomial degrees in the loop refer only to the dynamical coordinates. Alternatively, repeat the calculation at nearby parameter values while keeping phase gauges and mode labels consistent. The latter approach needs numerical convergence checks for the resulting derivatives.

The following steps check both the conjugacy and the form of the retained dynamics. Invariant-torus optics and tune derivatives are then extracted when the retained Hamiltonian depends only on actions.

```

N = current
C = T o L o B
check_coefficients(M o C - C o N, through_degree=d)
check_coefficients(N - R o E(Z), through_degree=d)
check_symplecticity(C, N, through_available_order=d - 1)

if only_action_terms_are_retained:
    H_NF = dot(mu, J) + Z
    tunes = gradient(H_NF, J) / (2*pi)
    chromaticities = derivatives(tunes, delta0, at=0)
    knob_responses = derivatives(tunes, lambda, at=lambda_star)
    for observation_location s:
        C_s = transport_map(s, from=s0) o C
        X = fourier_coefficients_of_embedding(C_s)
        beta_x_mode_j = 2 * abs(X[unit_harmonic(j)])**2 / J[j]

return C, N, Z, observables_if_nonresonant

```

4.9 Validity and Practical Checks

A finite-order normal form describes a neighbourhood of the chosen closed orbit. Its useful amplitude range depends on the nonlinear strengths, omitted orders, and resonance denominators. Formal existence of the series does not guarantee convergence. Under the additional nondegeneracy and nonresonance assumptions of KAM theory, many sufficiently small invariant tori survive a small Hamiltonian perturbation, but this does not imply that every action in a finite domain labels an exact torus.

Several checks directly test what the calculation claims. The conjugacy residual $\mathcal{M} \circ \mathcal{C} - \mathcal{C} \circ \mathcal{N}$ should have no coefficients through the retained coordinate degree. The computed maps should satisfy the symplectic Jacobian condition through the order available after differentiating the Taylor map. For an action-only normal form, track particles launched through $\mathcal{C}$ and compare measured tunes, harmonic amplitudes, and the variation of I_j with the predicted quantities. Increasing the map order or decreasing the actions should reduce discrepancies where the expansion is useful.

Near a separatrix or in a chaotic region, a single quasiperiodic torus embedding and its amplitude-dependent beta are no longer an adequate description. Dynamic aperture and long-term stability therefore require tracking and other diagnostics in addi-

tion to local normal forms. The practical value of the normal form is that it explains the local frequency shifts, orbit deformations, and resonances in one consistent canonical construction.

Chapter 5

Generalised TPSAs

5.1 TPSAs

Taylor series in one dimension:

$$f(x) = \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(a) (x - a)^k. \quad (5.1)$$

Taylor series in n dimensions:

$$\begin{aligned} f(\mathbf{a} + \Delta \mathbf{x}) &= \sum_{k=0}^{\infty} \frac{1}{k!} \partial_{i_1} \partial_{i_2} \cdots \partial_{i_k} f(\mathbf{a}) \Delta x^{i_1} \Delta x^{i_2} \cdots \Delta x^{i_k} \quad i_j = 1, \dots, n \\ &= \sum_{k=0}^{\infty} \frac{1}{k!} \left(\Delta x^i \partial_i \right)^k f(\mathbf{a}) \\ &= \sum_{|\alpha| \geq 0} \frac{1}{\alpha!} \Delta x^\alpha \partial^\alpha f(\mathbf{a}) \\ \alpha &= (\alpha_1, \alpha_2, \dots, \alpha_n) \\ \alpha! &= \prod \alpha_j! \\ \Delta x^\alpha &= \Delta x_1^{\alpha_1} \Delta x_2^{\alpha_2} \cdots \Delta x_n^{\alpha_n}. \end{aligned}$$

Let

$$\mathbf{r} = (x, p_x, y, p_y, \zeta, \delta)^T. \quad (5.2)$$

A six-dimensional Taylor map is a vector-valued map

$$\mathcal{M}(\mathbf{r}) = (\mathcal{M}_1(\mathbf{r}), \dots, \mathcal{M}_6(\mathbf{r}))^T, \quad (5.3)$$

with each component represented by a TPSA.¹ Thus a TPSA is a scalar truncated multivariate power series; a phase-space Taylor map is a vector of such TPSAs.

This means that each component K of the map is a polynomial of order N :

$$\mathcal{M}^K = \sum_{i+j+k+l+m+n \leq N} c_{ijklmn} x^i p_x^j y^k p_y^l \zeta^m \delta^n. \quad (5.4)$$

¹The longitudinal canonical coordinates are often chosen differently.

Practically, any TPSA can be represented by a list of powers and associated coefficients for each monomial, like so:

$$-0.376xy^3 \Rightarrow (1,0,3,0,0,0) : -0.376, \quad (5.5)$$

$$\begin{aligned} 0.8x + 0.97p_x - 0.0017x\delta &\Rightarrow (1,0,0,0,0,0) : 0.8 \\ &\quad (0,1,0,0,0,0) : 0.97 \\ &\quad (1,0,0,0,0,1) : -0.0017, \end{aligned} \quad (5.6)$$

etc. Any operation on a map is then just a matter of bookkeeping of its polynomial's powers and combining the relevant coefficients.

For instance, a derivative acts on the powers as follows:

$$\begin{aligned} \partial_x [(i,j,k,l,m,n) : c] &= (i-1,j,k,l,m,n) : ic \\ \partial_{p_x} [(i,j,k,l,m,n) : c] &= (i,j-1,k,l,m,n) : jc \\ \partial_y [(i,j,k,l,m,n) : c] &= (i,j,k-1,l,m,n) : kc \\ &\dots \\ \partial_x^3 \partial_{p_x} [(i,j,k,l,m,n) : c] &= (i-3,j-1,k,l,m,n) : i(i-1)(i-2)jc \quad \text{if } i \geq 3 \text{ else } 0 \\ &\dots \end{aligned} \quad (5.7)$$

5.2 Jacobian

The Jacobian matrix of a map is given by

$$\mathcal{J}_{ij} = \partial_j \mathcal{M}_i, \quad (5.8)$$

which is straightforward to calculate for TPSAs using Eq. 5.7.

The **Jacobian determinant** is the determinant of the Jacobian matrix:

$$\mathcal{J} = |\mathcal{J}_{ij}|. \quad (5.9)$$

The Jacobian determinant alone is not a sufficient symplecticity test in more than one degree of freedom. For canonical coordinates, a Taylor map is symplectic when its Jacobian matrix satisfies

$$\mathcal{J}(\mathbf{r})^T S \mathcal{J}(\mathbf{r}) = S, \quad (5.10)$$

through the polynomial order available after differentiation. In particular, $\det \mathcal{J} = 1$ is a necessary consequence, but not the full condition.

Bibliography

- [1] Alex J. Dragt and John M. Finn. Lie series and invariant functions for analytic symplectic maps. *Journal of Mathematical Physics*, 17(12):2215–2227, 1976.
- [2] Alex J. Dragt and Etienne Forest. Computation of nonlinear behavior of hamiltonian systems using lie algebraic methods. *Journal of Mathematical Physics*, 24(12):2734–2744, 1983.
- [3] Étienne Forest. *Beam Dynamics: A New Attitude and Framework*, volume 8 of *The Physics and Technology of Particle and Photon Beams*. Hardwood Academic / CRC Press, Amsterdam, The Netherlands, 1998.
- [4] Etienne Forest. *From Tracking Code to Analysis: Generalised Courant-Snyder Theory for Any Accelerator Model*. Springer, 2016.
- [5] Andrzej Wolski. Alternative approach to general coupled linear optics. *Physical Review Special Topics – Accelerators and Beams*, 9:024001, 2006.